

Data Science

CS301

Modeling: Formal Basics

Week 09

Fall 2025

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Are you here today?!

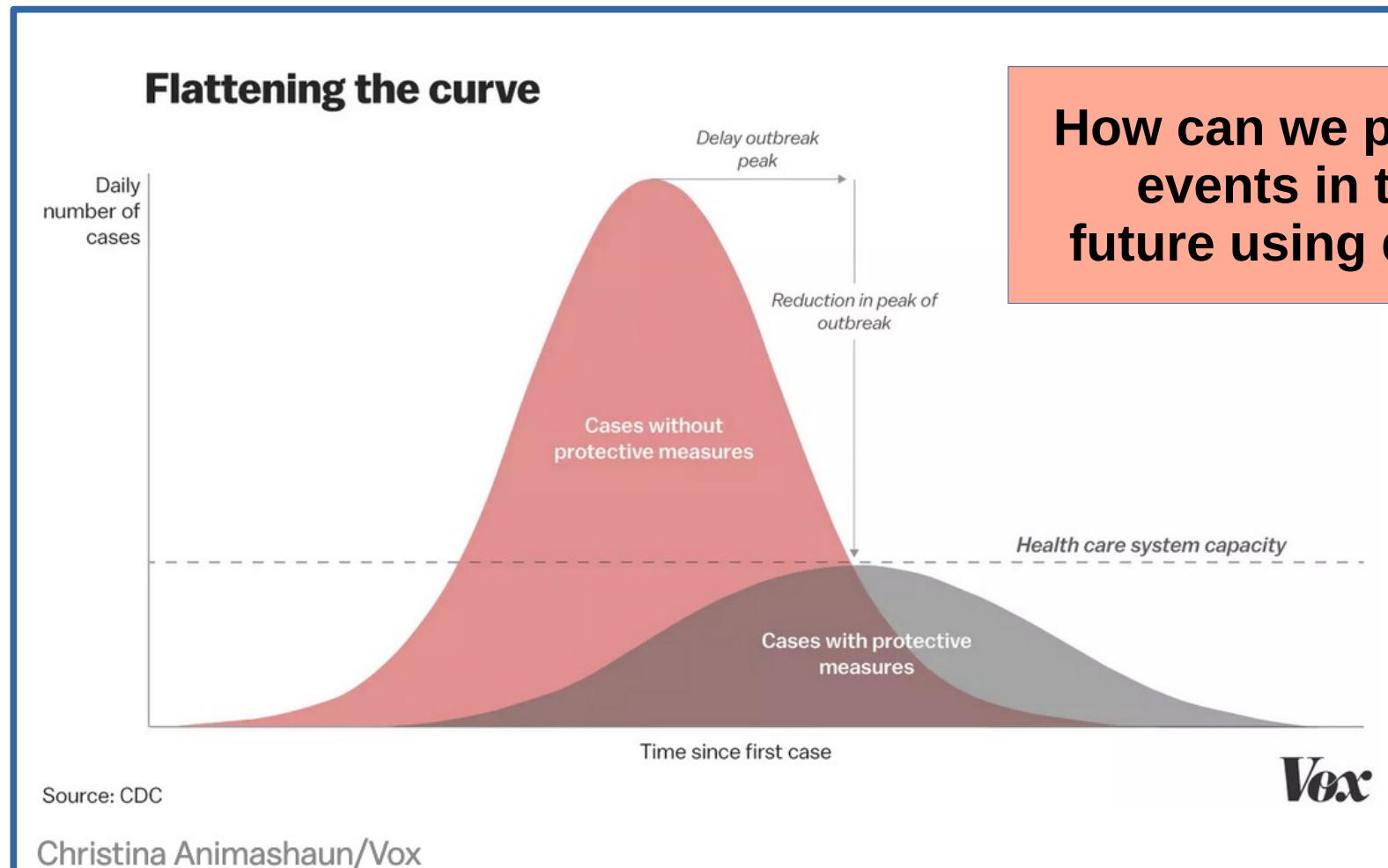
ATTENDANCE

Click, Complete:

<https://forms.gle/K8yZJNJTU8soNQX27>

Trends From Plots

- How canceled events and self-quarantines save lives, in one chart

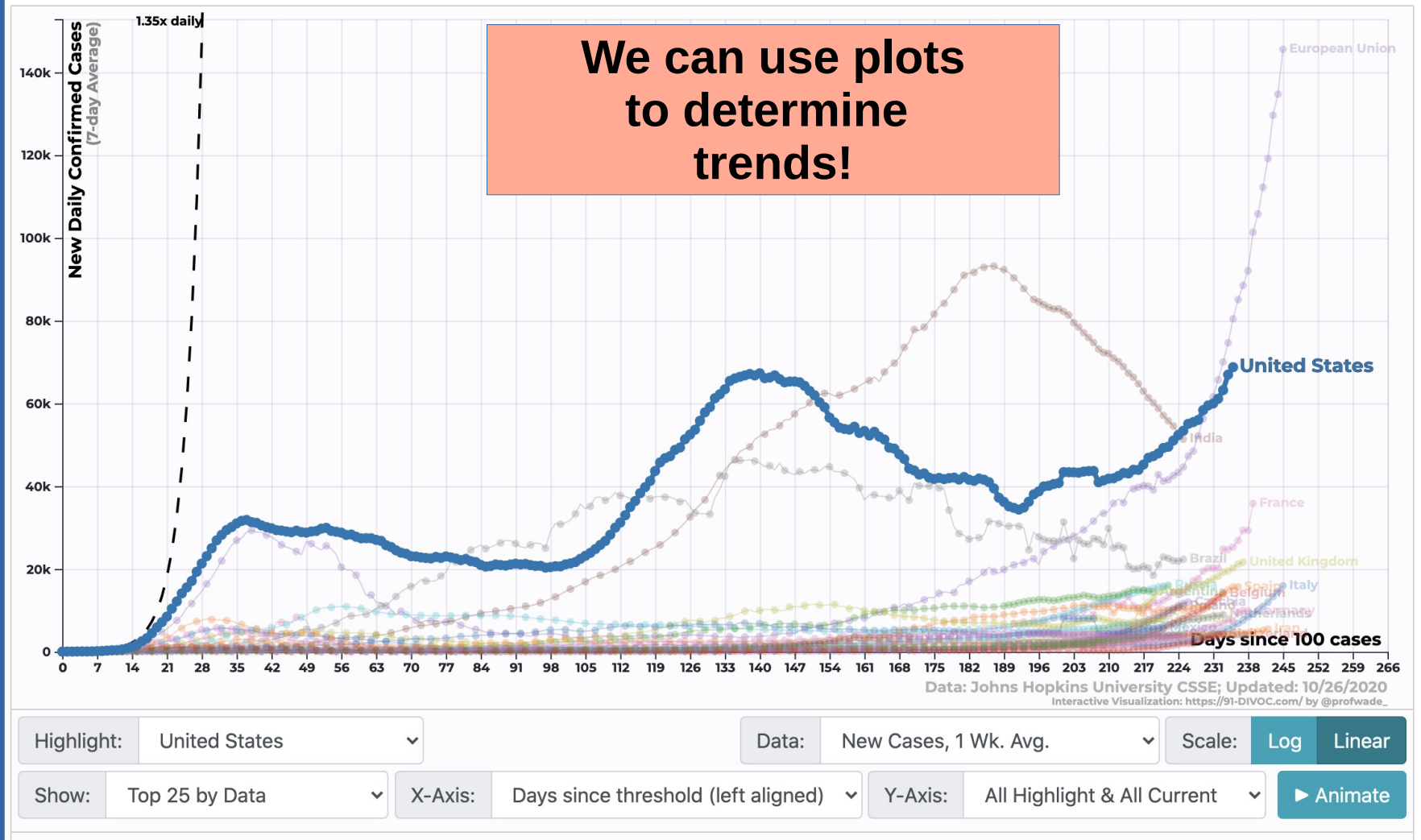


How can we predict events in the future using data?



Interactive Plots: Covid-19 Cases

New Confirmed COVID-19 Cases per Day



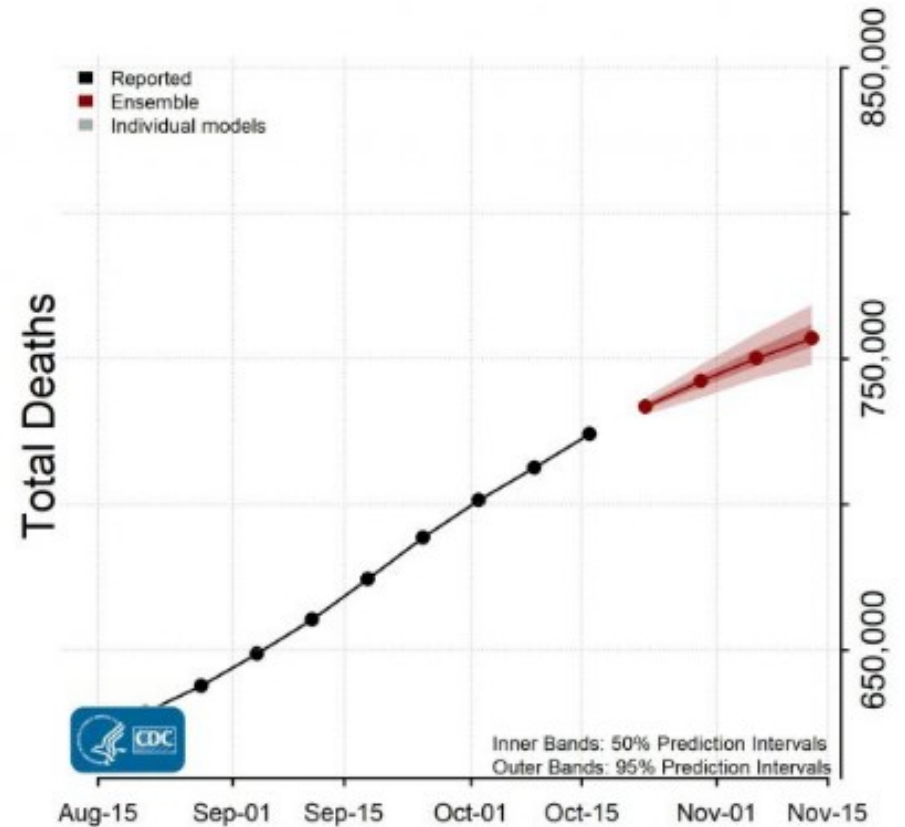
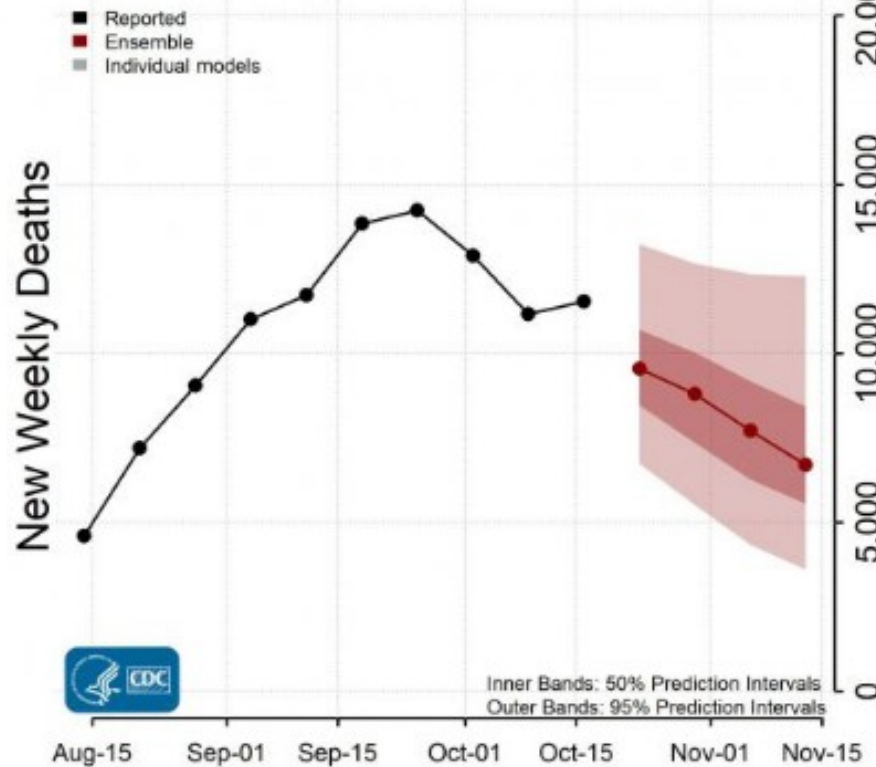


Predicting With Plots

National Forecast

More types of trends!

National Forecast Combined Forecast





The “Stuff” of Knowledge

- How do we use data to explain or argue something?
- How can we use data to make **predictions**?





Modeling Basics

- What are models?
 - Data does not provide much insight unless something can be learned from it
 - Prediction comes from the ability to see meaningful trends in data
- Let's talk about...
 - How to extract some meaning from your data
 - How to make predictions based on training by data



Types of Models (i)

- **Support Vector Machines**
 - Supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis.
- **Generalized Linear Models**
 - Flexible generalization of ordinary linear regression that allows for response variables that have error distribution models other than a normal distribution
- **Generalized additive models**
 - Generalized linear model in which the linear predictor depends linearly on unknown smooth functions of some predictor variables, and interest focuses on inference about these smooth functions



Types of Models (ii)

- **Linear Regression**

- Linear approach for modeling the relationship between a scalar dependent variable y and one or more explanatory variables (or independent variables) denoted X
- *(we have sort-of begun this study already: lines in scatter plots)*

- **LOESS Regression**

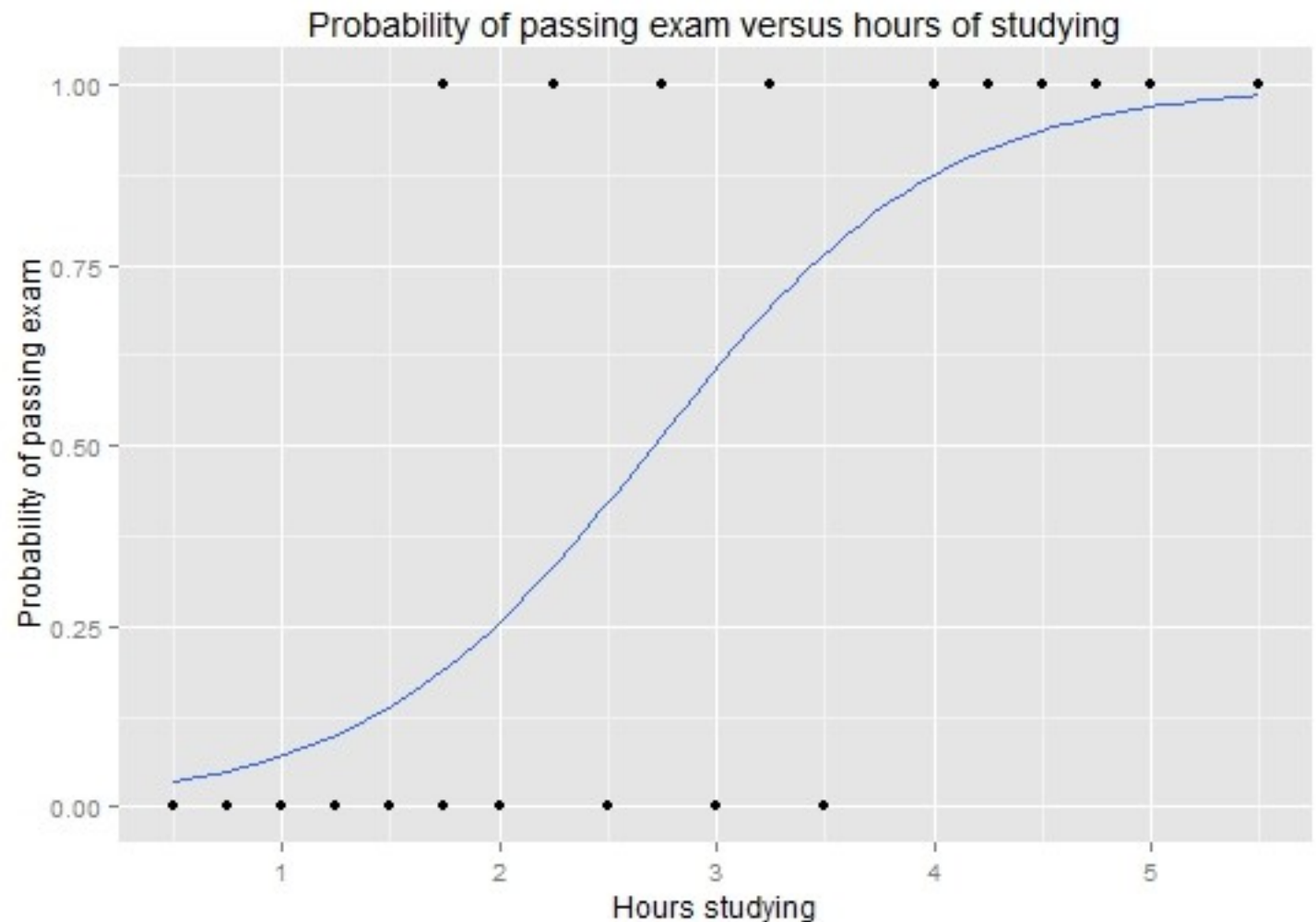
- Combining much of the simplicity of linear least squares regression, but building with the flexibility of nonlinear regression.



Types of Models (iii)

Logistic Regression

- Models where the dependent variable is categorical (i.e., 0's or 1's as factors)





Answering Complex Questions

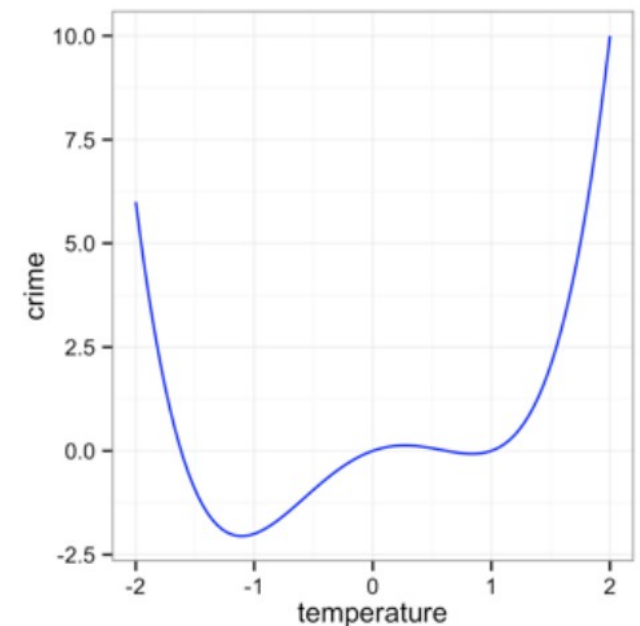
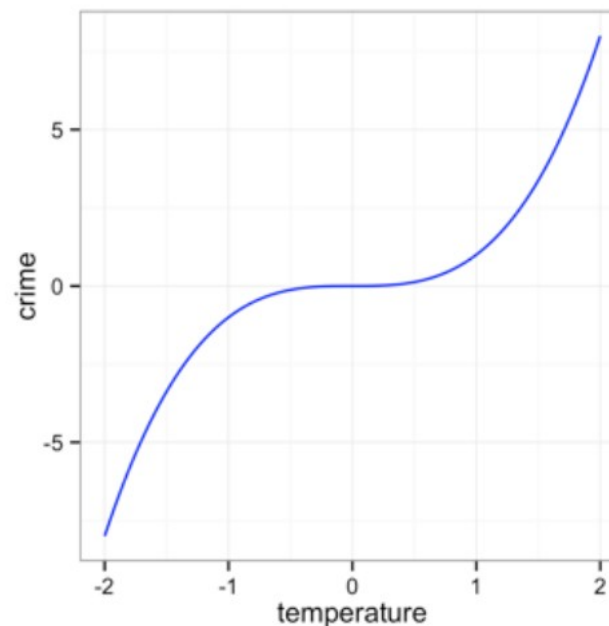
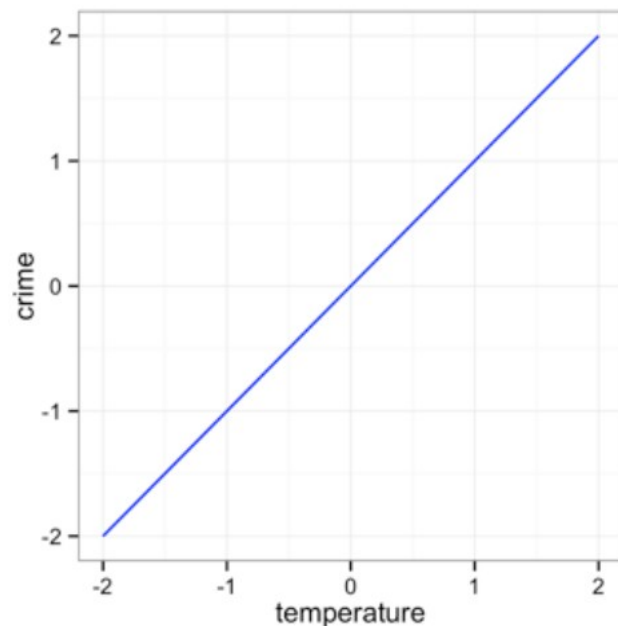
- **Modeling:** We employ a computational framework based-on historical data trends.
- **Prediction:** We apply changes to a variable to see what happens to other variables ...

What if...



Functions: the *stuff* behind the models

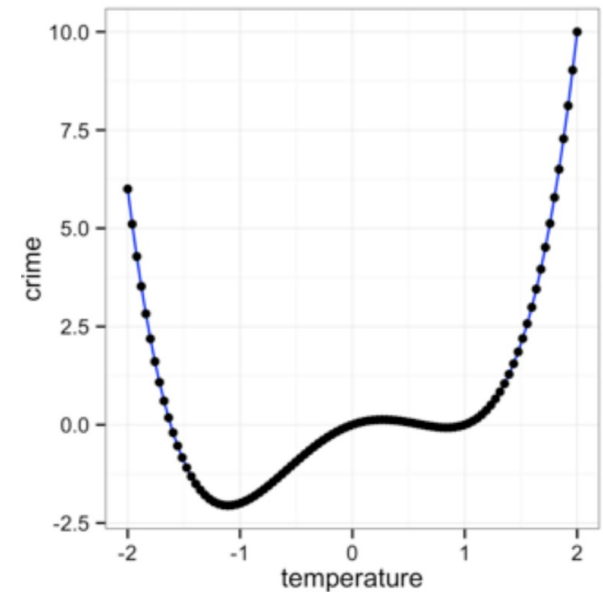
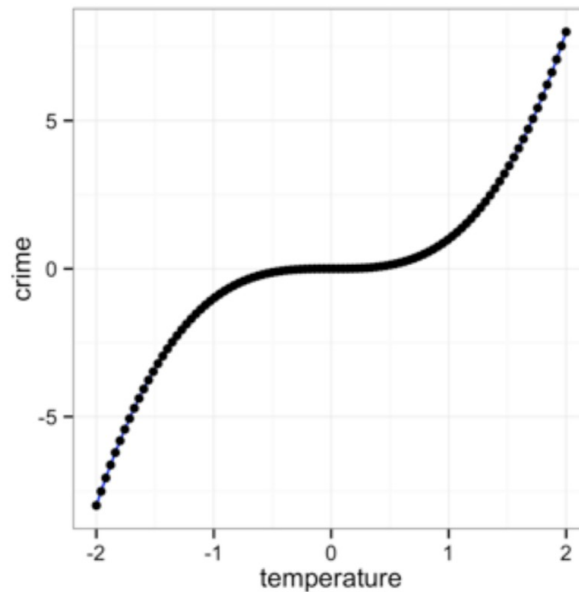
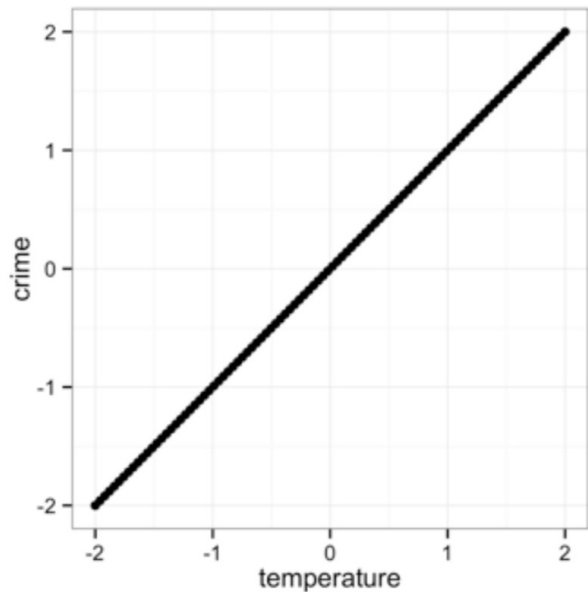
- Ideally, functions are mathematical descriptions of relationships





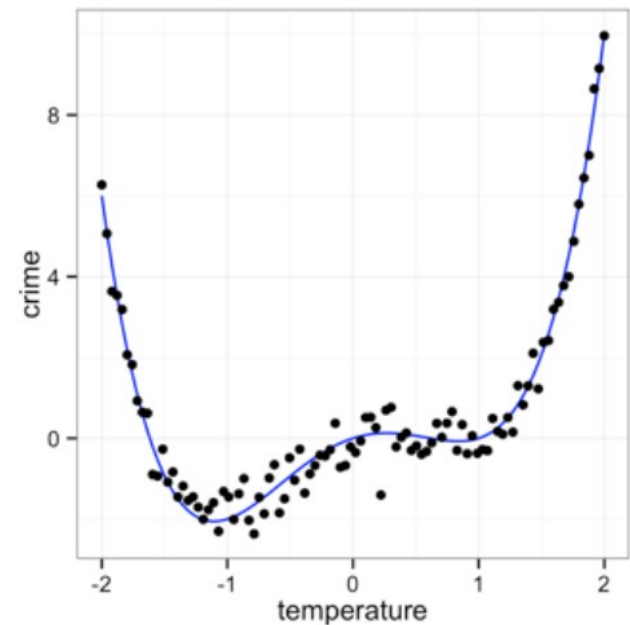
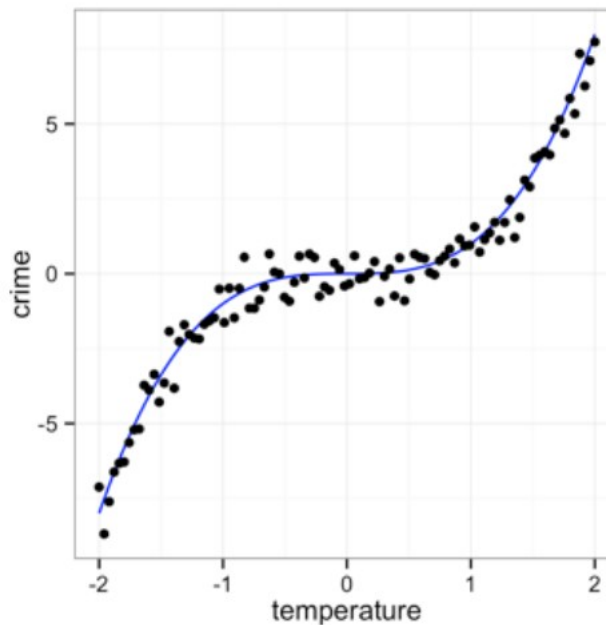
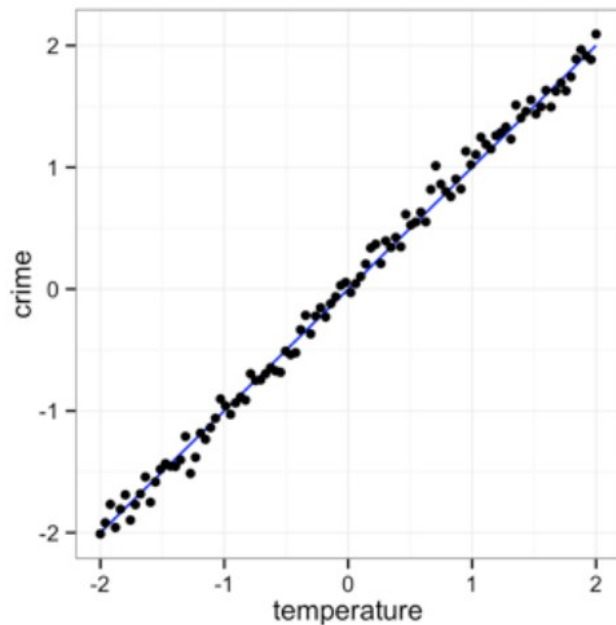
Functions: the *stuff* behind the models

- If one variable *completely* determines another, every (x, y) data point will fall on the function's line.



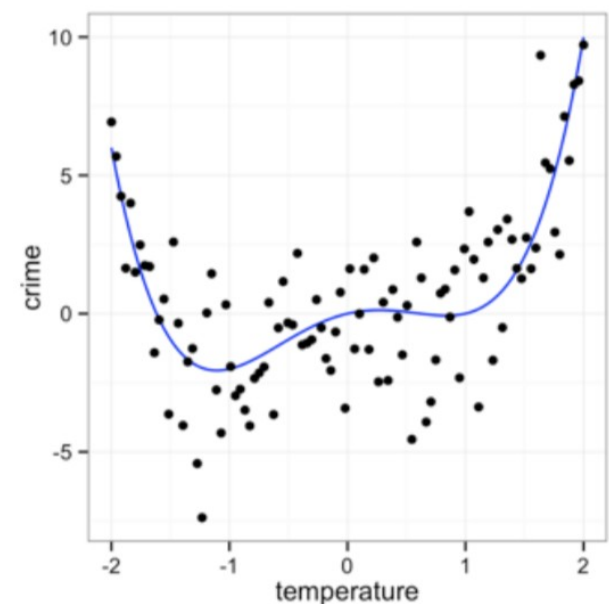
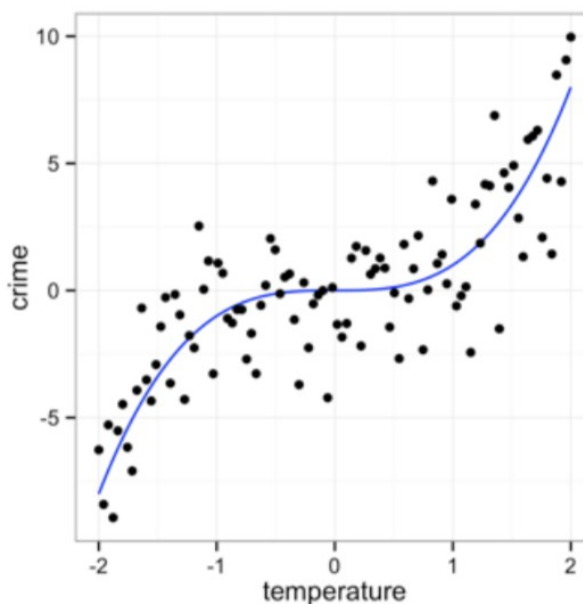
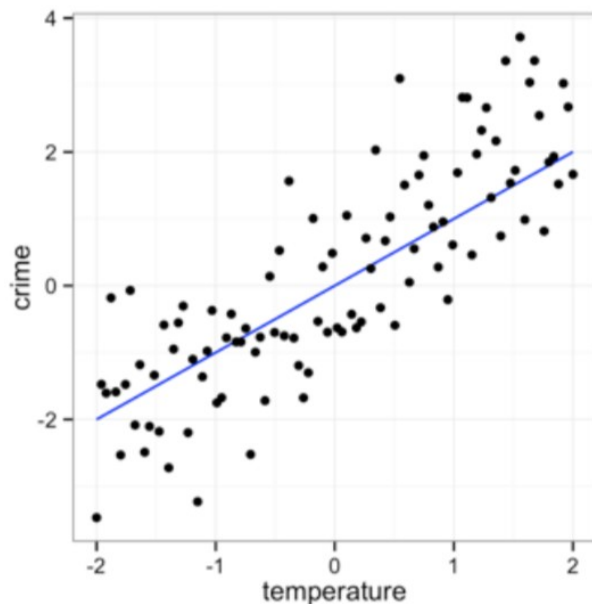
Relationships Between Variables are Messy

- This is what real data looks like on a good day!



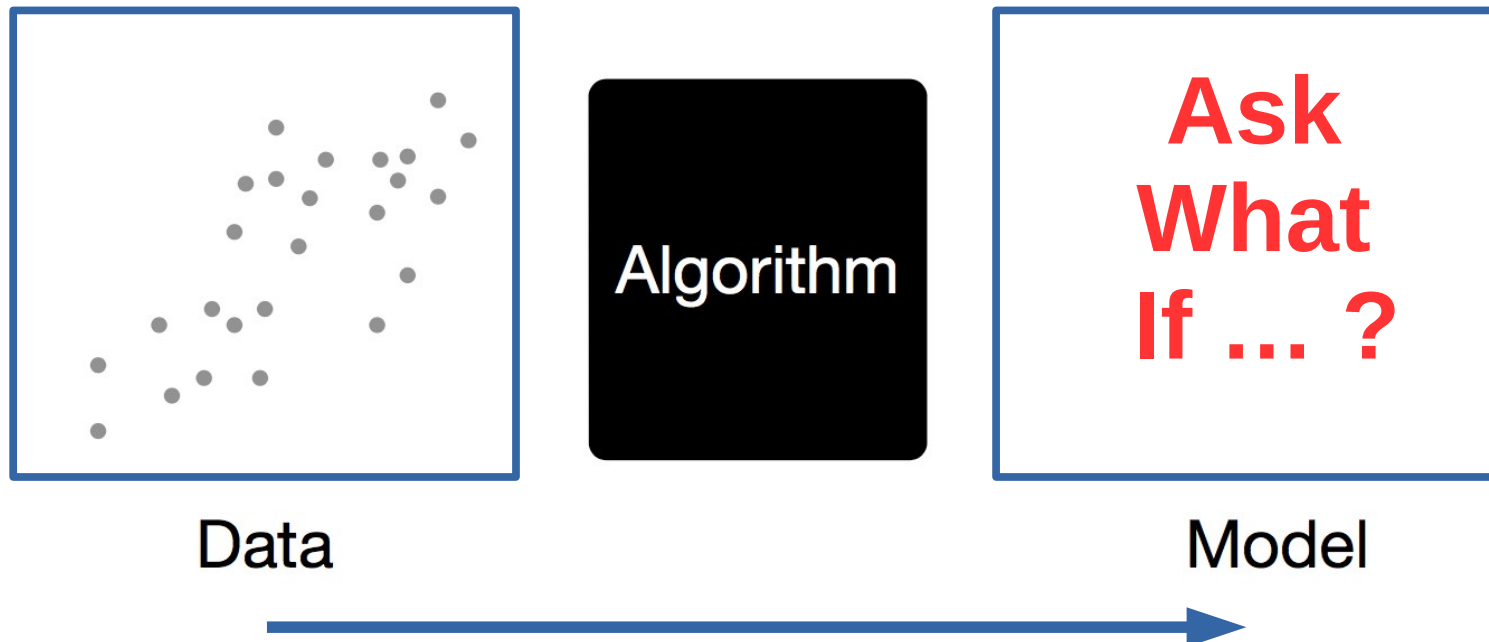
Relationships Between Variables

- If the actual relationship is affected by other variables, data points may not fall directly on the function-line.
- **Noise:** The greater the effect of other variables, the weaker the relationship. This is normally the situation with real data.



So, A Model, Then?

- Noise is what we get in data when not every point does *what it is supposed to do*.
- Modeling *attempts* to *more-correctly* identify functional relationships in noisy data.





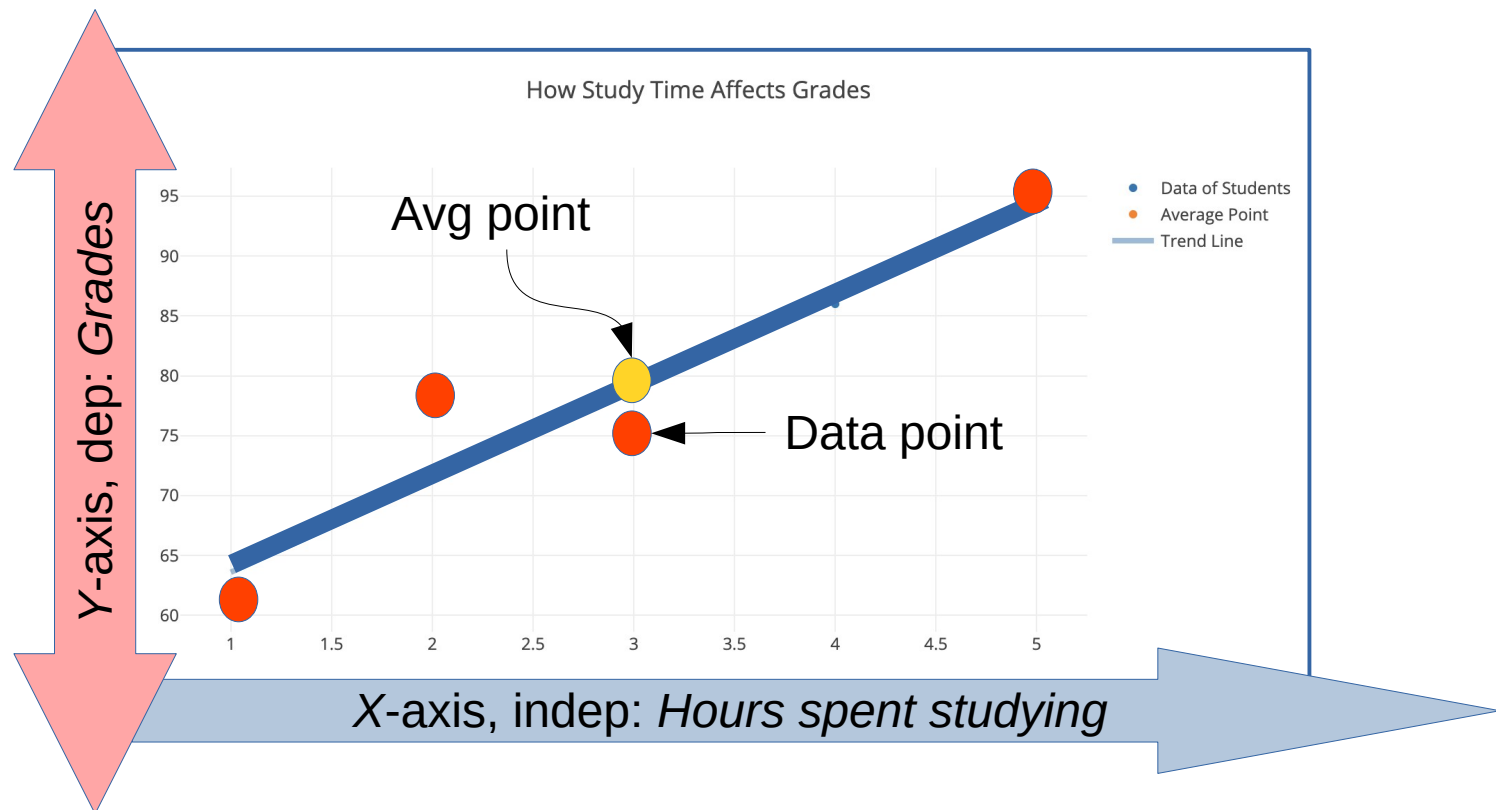
Let's Talk Linear Models

- **Linear regression:** How much do/does my **independent variable(s)** influence my **dependent variable(s)**?
- As one variable climbs, does the other also climb (decline) at some *predicable* rate?
- Can I impose some value into my model to determine a *what-if* type of question which is firmly based on my data?



Variables?

- **Independent variable:** a variable (often denoted by x) whose variation does not depend on that of another (i.e., time).
- **Dependent variables:** a variables (often denoted by y) that depends, by some law or rule (e.g., by a mathematical function), on the values of other variables (i.e., grades).
- Example: <https://chart-studio.plotly.com/~bchapman27/73.embed>





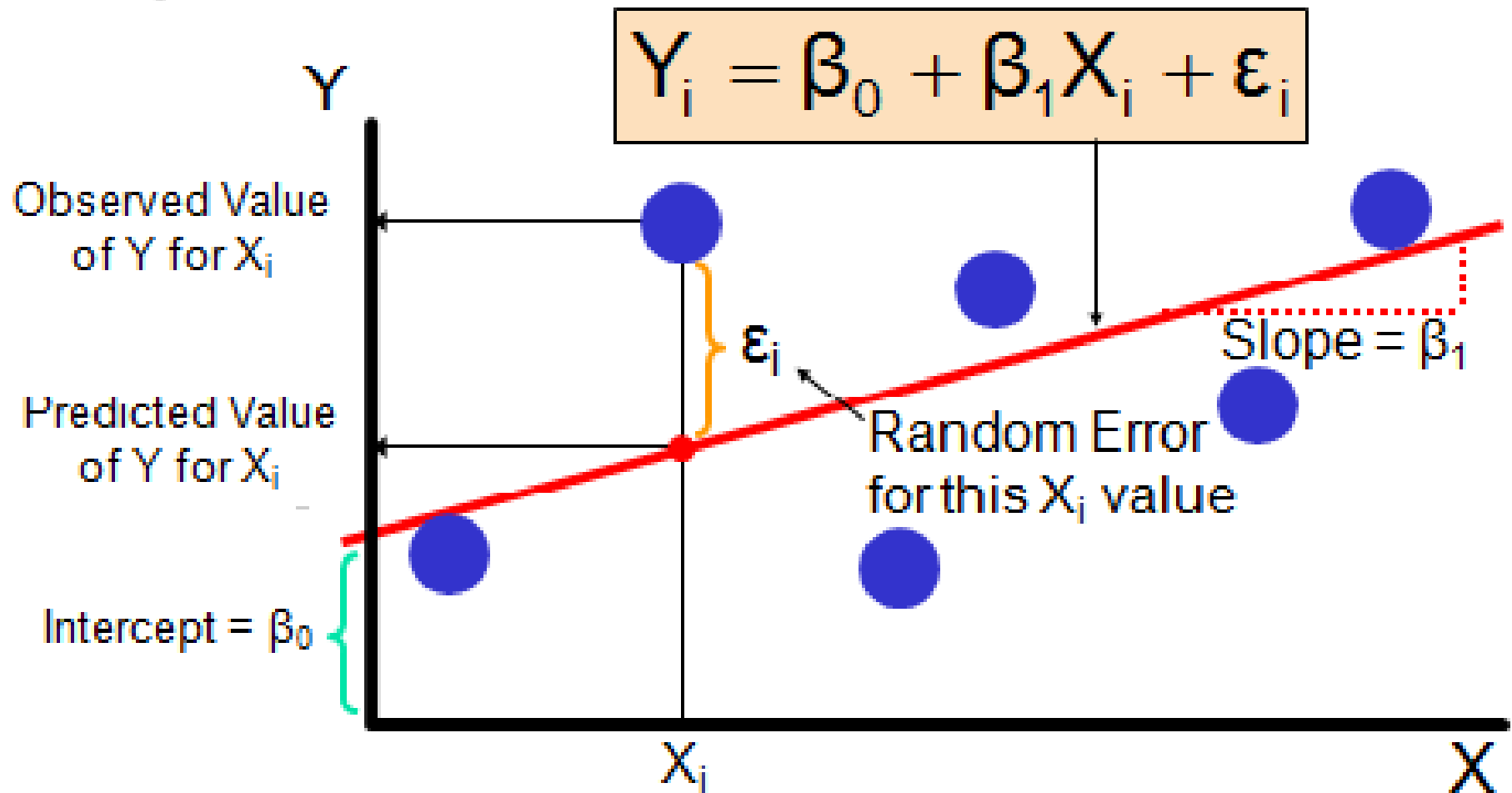
Let's Talk Linear Models

- Linear regression (formally) is a function that draws a line from data
- The function $f(x)$ has the form:

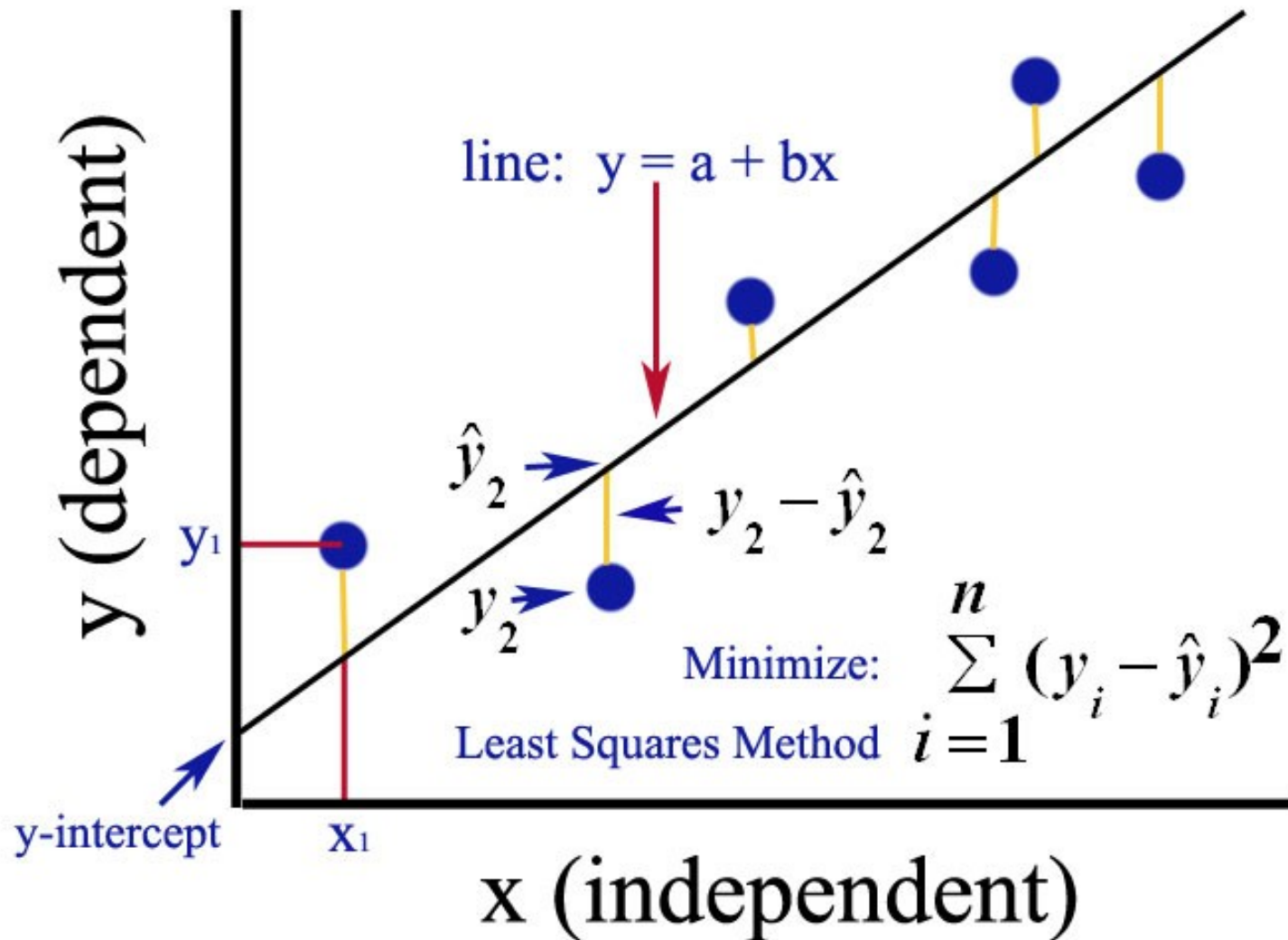
$$f(x) = \alpha + \beta x + \epsilon$$

- **alpha: intercept**
 - **Beta: a weighted slope**
 - **Epsilon: account for the error**
- *Note: This $f(x)$ will be a straight line for x*

Just a Formula For a Line!



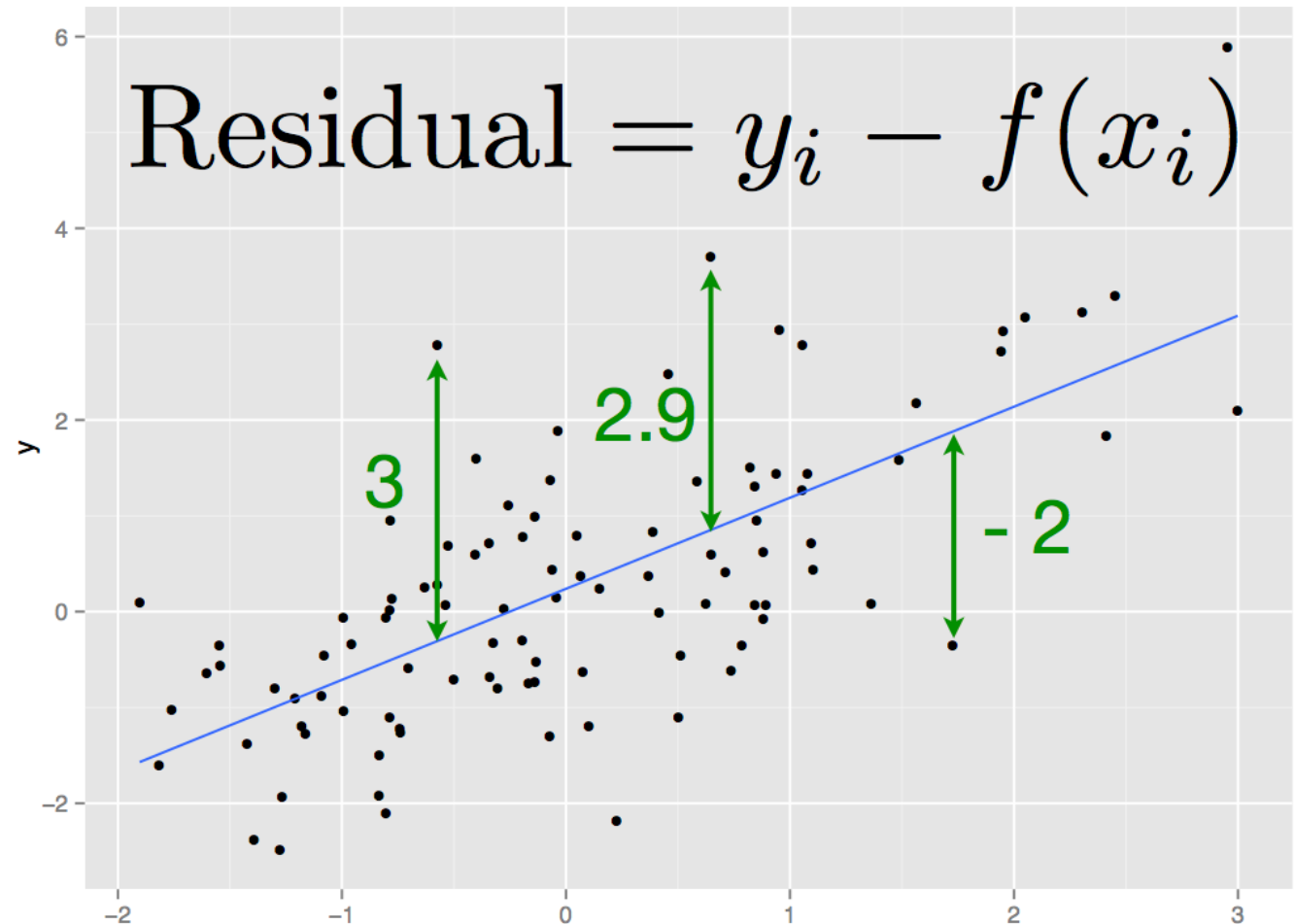
Another Linear Model





Residuals

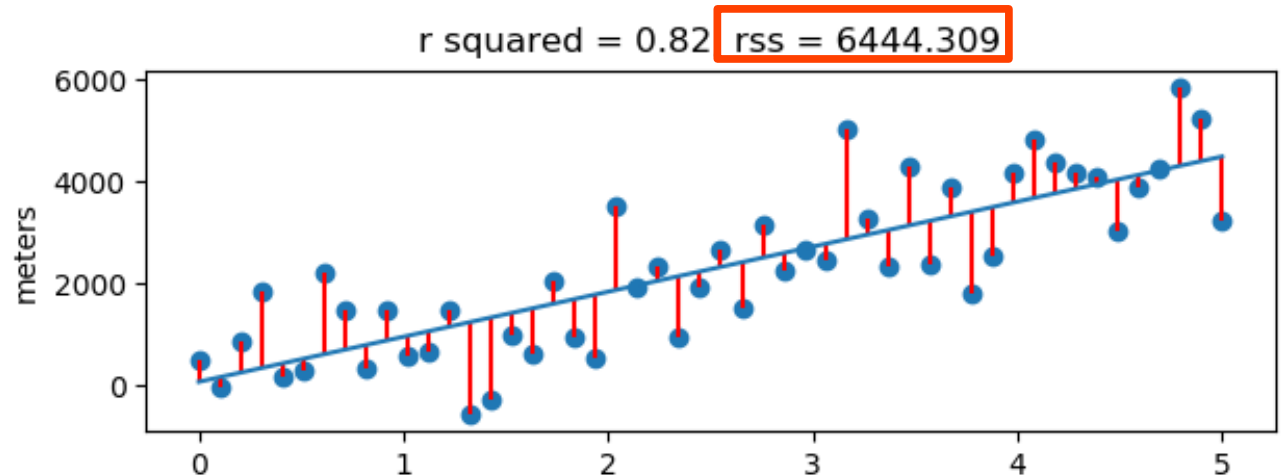
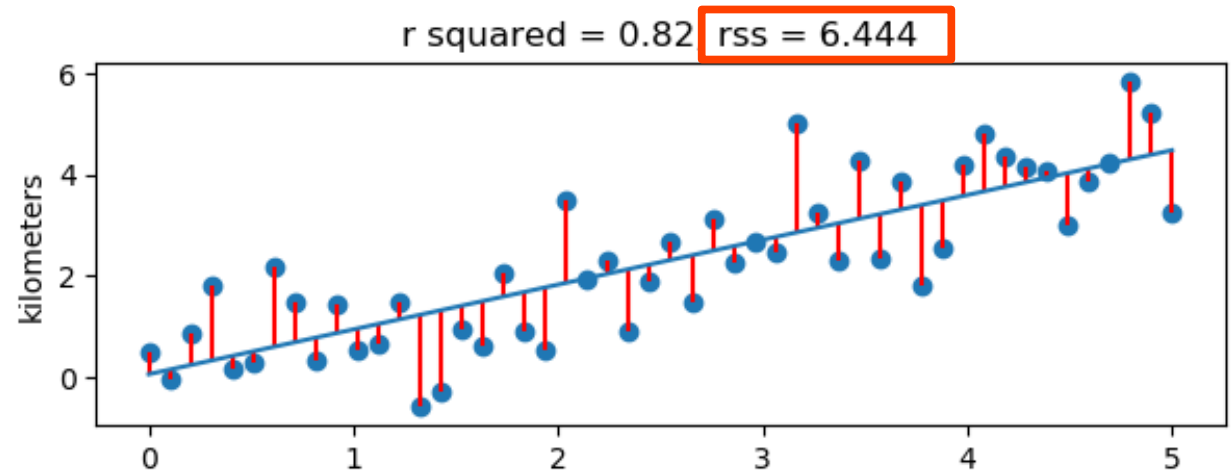
- A residual is the difference between the actual value and the value predicted by the model ($y - \hat{y}$) for any given point.
- A least-squares regression model minimizes the sum of the squared residuals





A Difference Between the Expected and the Actual Data

- The *Residual Sum of Squares* is a measure of the discrepancy between the data and an estimation model.
- A **small RSS** value indicates a **tight fit** of the model to the data; better prediction





Types of Questions to Address With Data

Q1: Is crime influenced by yearly temperature?

File: crime.csv



Q2: What influence is there on earning potential and personal height?

File: wages.csv

Crime Data Set



- Is there a relationship between *crime* and *temperature*?
State statistics from 2009.

```
rm(list = ls()) # remove old vars in memory
library(tidyverse)
# open the crime dataset from the data.
c <- file.choose() # set the filename
crime <- read.csv(c) # load and read the data.
```



Crime Data Set

```
View(crime) #or  
tibble::as_tibble(crime) # dataframe
```

	state <chr>	abbr <chr>	low <int>	murder <dbl>	tc2009 <dbl>
1	Alabama	AL	-27	7.1	4337.5
2	Alaska	AK	-80	3.2	3567.1
3	Arizona	AZ	-40	5.5	3725.2
4	Arkansas	AR	-29	6.3	4415.4
5	California	CA	-45	5.4	3201.6
6	Colorado	CO	-61	3.2	3024.5
7	Connecticut	CT	-32	3.0	2646.3
8	Delaware	DE	-17	4.6	3996.8
9	Florida	FL	-2	5.5	4453.7
10	Georgia	GA	-17	6.0	4180.6
...					

Yearly low temp

Murder rate

Training data



Exploratory Plots

#plot with general trend line

```
crime %>% ggplot(aes(x = low, y = tc2009)) +  
  geom_point(alpha = I(1/4)) + geom_smooth()
```

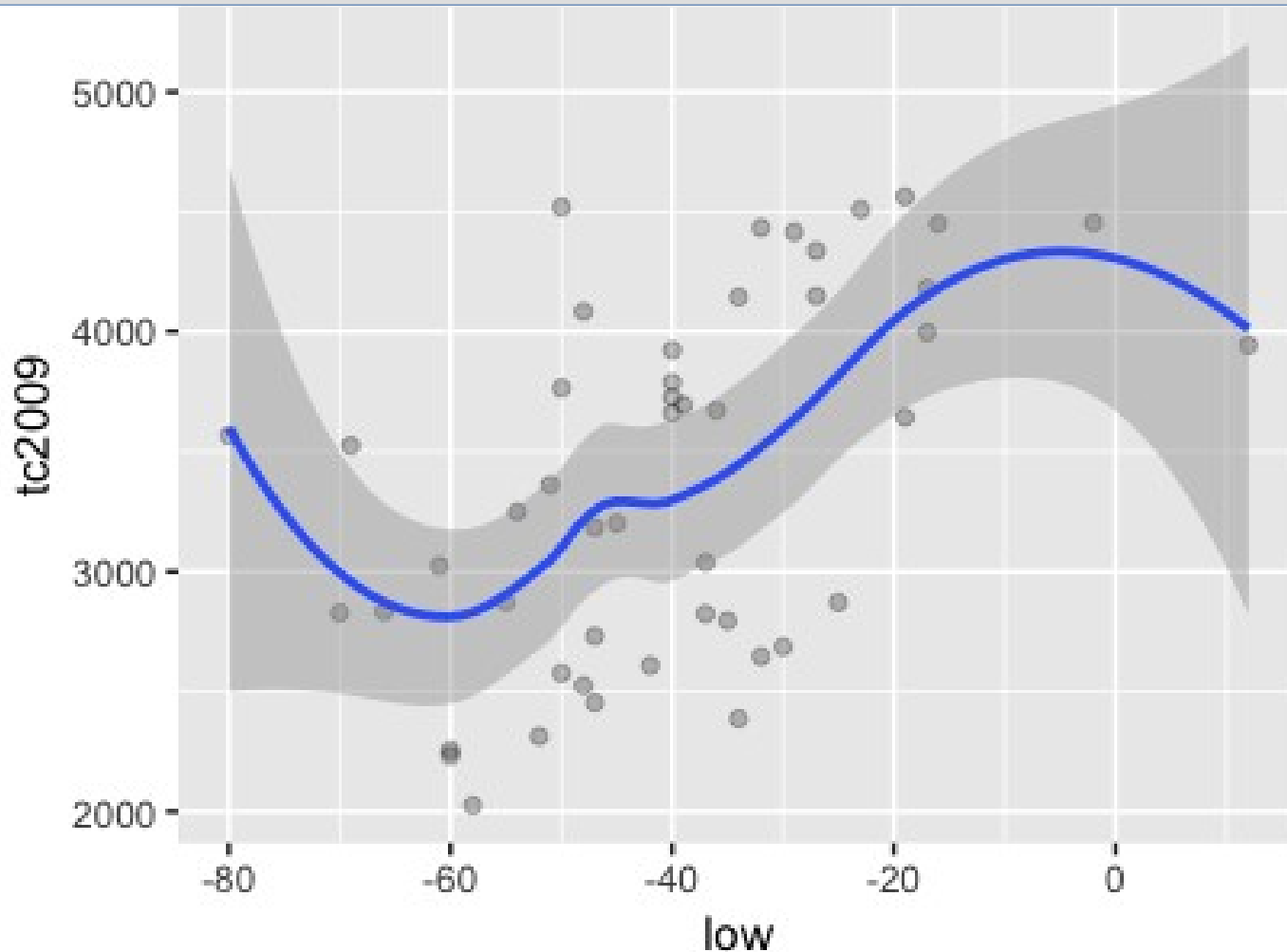
#plot with linear model line

```
crime %>% ggplot(aes(x = low, y = tc2009)) +  
  geom_point(alpha = I(1/4)) +  
  geom_smooth(method = lm)
```



No Model: Just General Trends

```
crime %>% ggplot(aes(x = low, y = tc2009)) +  
  geom_point(alpha = I(1/4)) + geom_smooth()
```

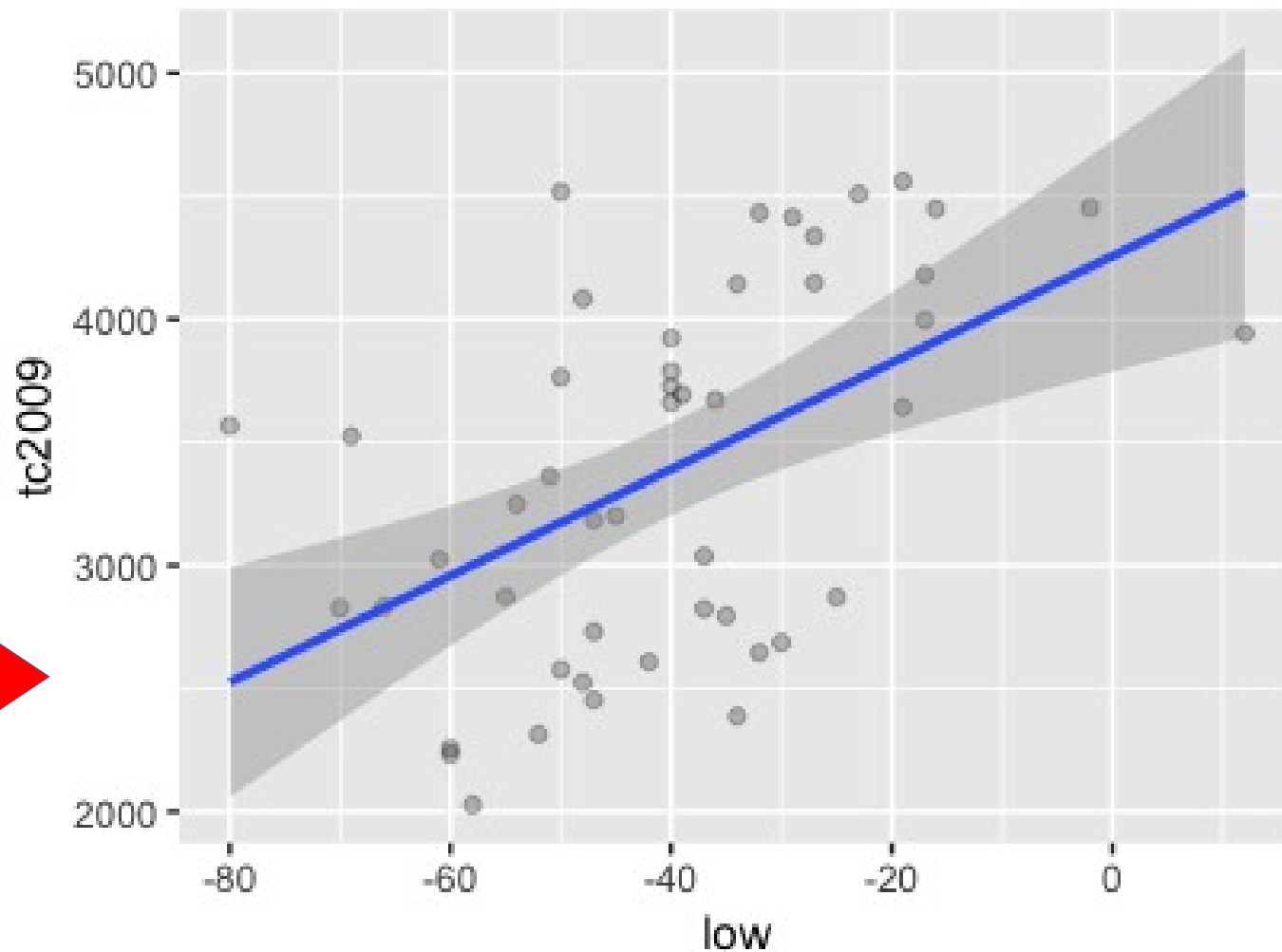


**This is
the
data's
general
pattern**



Linear Model: Predictions

```
crime %>% ggplot(aes(x = low, y = tc2009)) +  
  geom_point(alpha = I(1/4)) + geom_smooth(method = lm)
```



**The
linear
Model:
used
for
predictions**



The Linear Model

- How much does *low* (*indep Var*) influence *tc2009* (*dep Var*)
- Linear model syntax

lm

Model formula:
response ~ predictor(s)

data

```
mod <- lm(tc2009 ~ low, data = crime)
```



Syntax

- Formulas only need to include the response and predictor variables

$$y = f(x) = \alpha + \beta x + \epsilon$$

Syntax to Build the linear model:

$$y \sim x$$



Models Use Formulas

- R formulas are expressions built with “**~**” (*tilda*)

```
tc2009 ~ low
```

```
# Note: tc2009 is dependent variable and low  
is independent variable
```

```
class(tc2009 ~ low)
```

```
# gives: [1] "formula" meaning that is a line  
equation
```

- $tc2009 = f(low) = \beta * low + \epsilon$

Different Names, Same Meanings

response ~ explanatory

dependent ~ independent

outcome ~ predictors



Build Your Model!

```
mod <- lm(tc2009 ~ low, data = crime)
```

Dependent ~ independent

Call:

```
lm(formula = tc2009 ~ low, data = crime)
```

Coefficients:

(Intercept)	low
4256.86	21.65



Intercept and Coefficient

mod

```
> mod
```

```
Call:
```

```
lm(formula = tc2009 ~ low, data = crime)
```

```
Coefficients:
```

(Intercept)	low
4256.86	21.65

Coef

- Shows the model's coefficients (i.e., intercept, slopes)

```
coef(mod)
```

```
coefficients(mod)
```

```
# (Intercept)
```

```
# 4256.86158
```

 α

```
21.64725
```

 β

slope

low



Interpreting Models

Linear models are very easy to interpret

$$y = \alpha + \beta x + \epsilon$$

α is the expected value of y when x is 0.

β is the expected increase in y associated with a one unit increase in x



Coefficients

`coef(mod)`

`coefficients(mod)`

(Intercept) low

4256.86158 21.64725

The best estimate of
tc2009 for a state with low = -10 is

$$4256.86 + 21.6 * (-10) = 4040.86$$

$$(x,y) \leftarrow (-10, 4040.86)$$



Coefficient Calculator

This function is now my data!!

Based on our training using data,
if $x = -10$, then $y = 4040.86$

The best estimate of
tc2009 for a state with low = -10 is
 $4256.86 + 21.6 * (-10) = 4040.86$

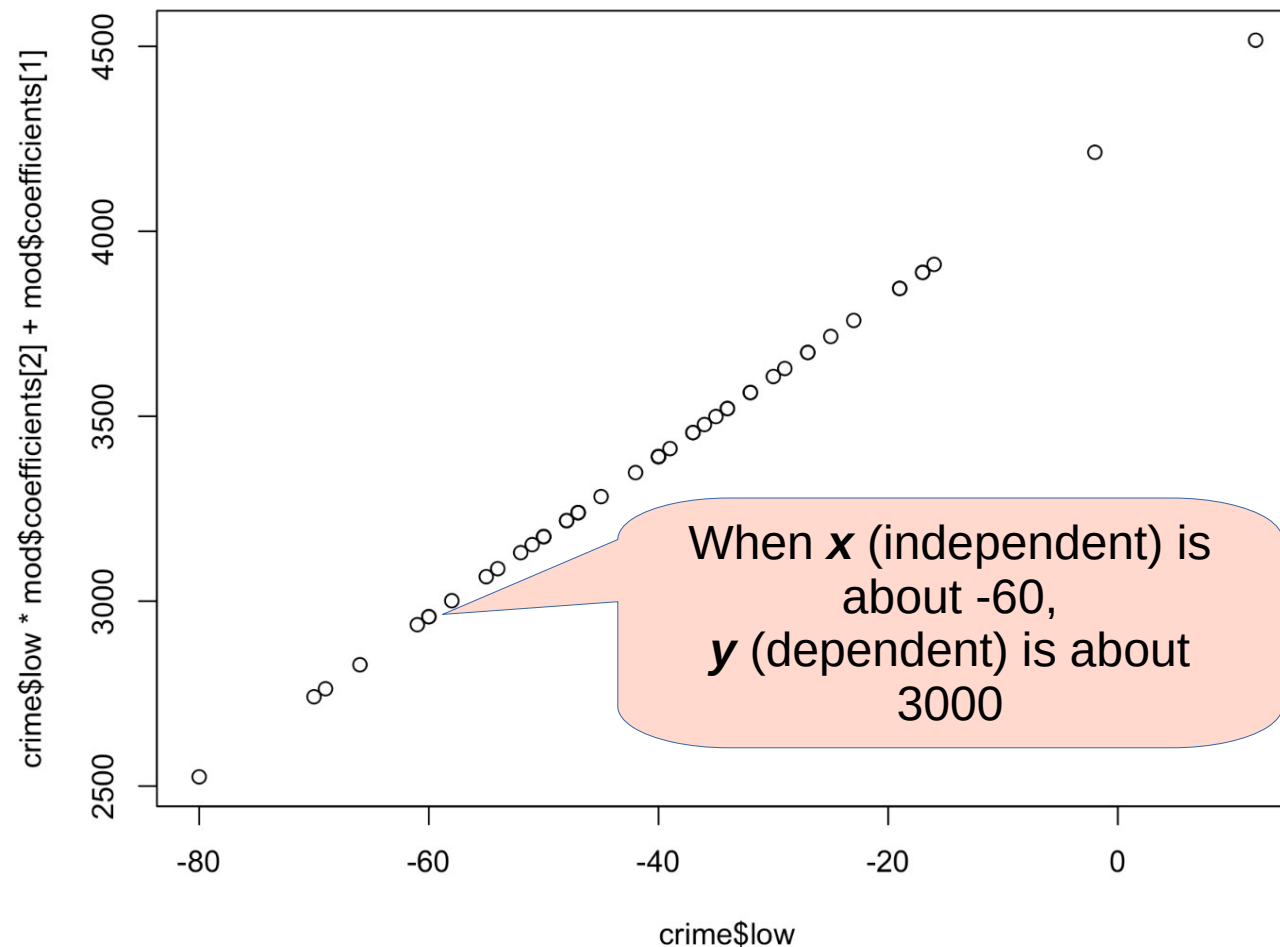
I can predict y ,
based on values of x !

**Due to
error,
there
may be
a slight
difference
between
expected
and
actual
values**



Forecasting Trends With a Simple Line

```
plot(crime$low, crime$low*mod$coefficients[2] + mod$coefficients[1] )
```





Coefficient Calculator Function for *mod*

```
# Function to compute estimated y value for an entered x  
value
```

```
tellMeY <- function(x_int){  
  
  cat("  intercept :",mod$coefficients[1] )  
  cat("\n  slope      :",mod$coefficients[2] )  
  y = mod$coefficients[1] + x_int * mod$coefficients[2]  
  cat("\n Model predicts y = ",y, "from x = ",x_int)  
}
```

```
# what if x = -10?  
tellMeY(-10) # note: x = -10
```

```
> # what if x = -10?  
> tellMeY(-10) # note: x = -10  
intercept : 4256.862  
slope      : 21.64725  
Model predicts y = 4040.389 from x = -10  
> |
```

$$\begin{aligned} y &= \text{intercept} + \text{slope} * x \\ &= \alpha + \beta x \\ &= b + mx \end{aligned}$$

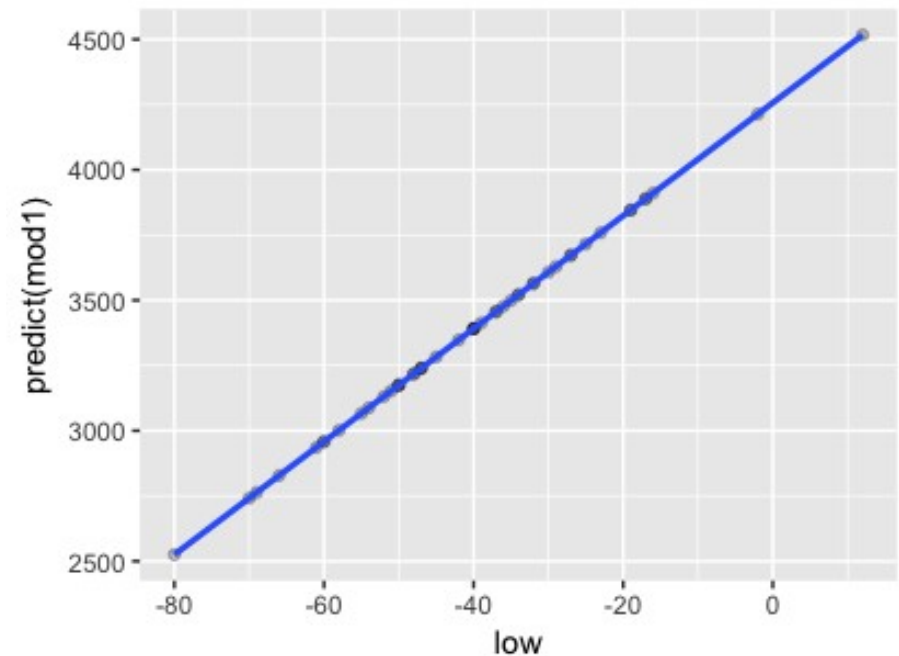
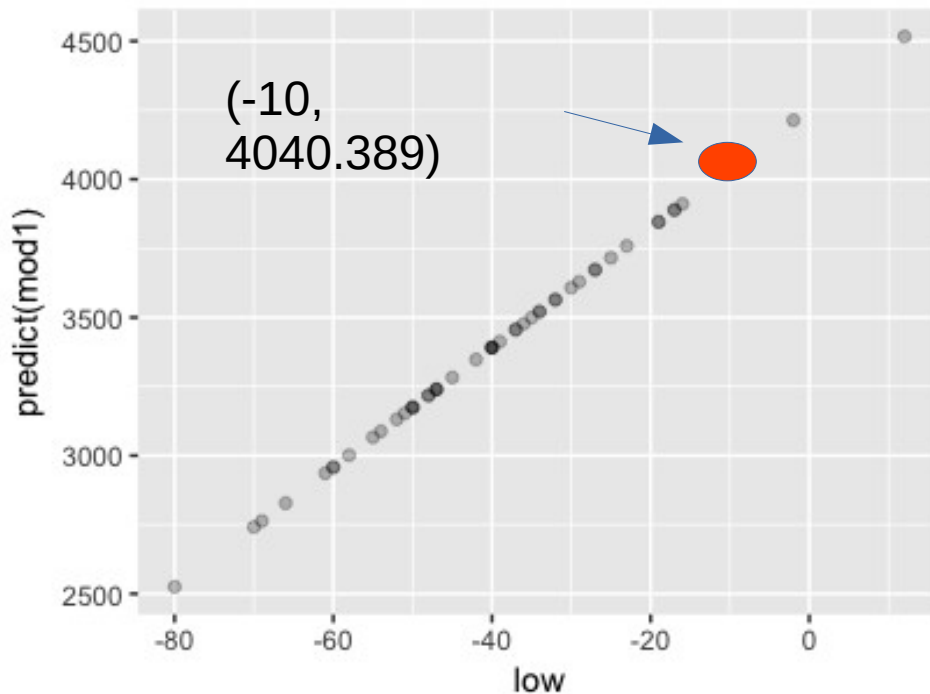


Forecasting with `predict()`

?predict

```
crime %>% ggplot(aes(x = low, y = predict(mod))) +  
  geom_point(alpha = I(1/4))
```

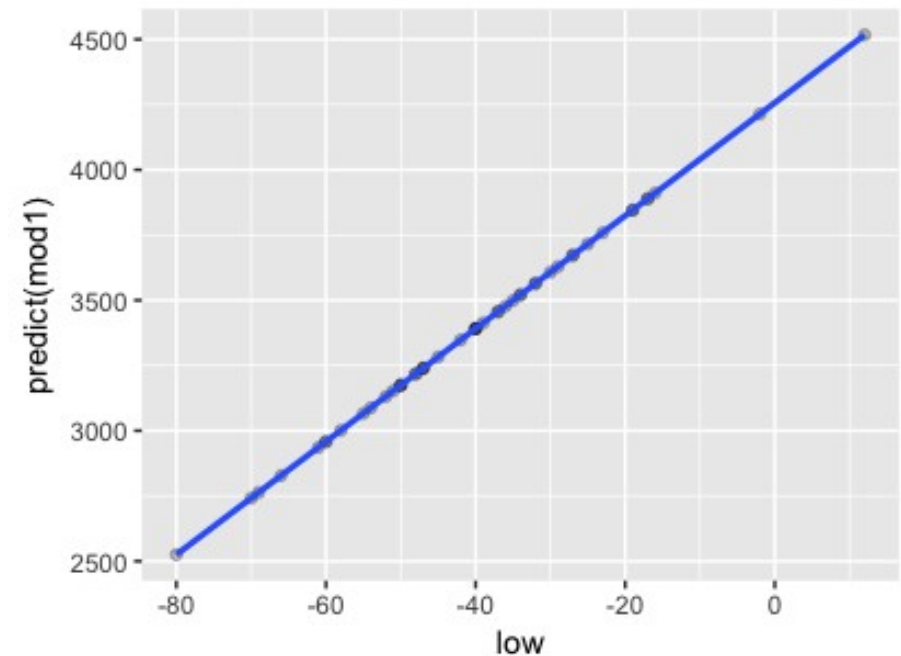
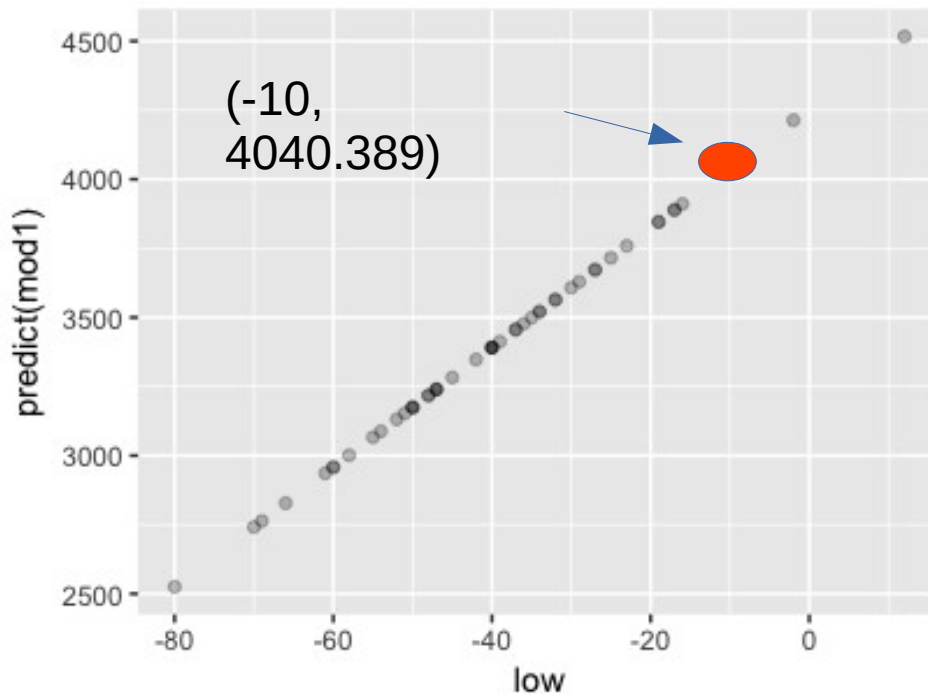
```
crime %>% ggplot(aes(x = low, y = predict(mod))) +  
  geom_point(alpha = I(1/4)) + geom_smooth()
```





What Conclusions to Draw?

After you have completed the analysis, what does the data indicate?





So, back to the Question... ?

**Q1: Is crime influenced
by yearly temperature?**

File: crime.csv



*A: The data and its trained model
suggest that there is a positive
correlation between crime and
temperature in the US*



Aside: intercept terms

R includes an intercept term in each model by default

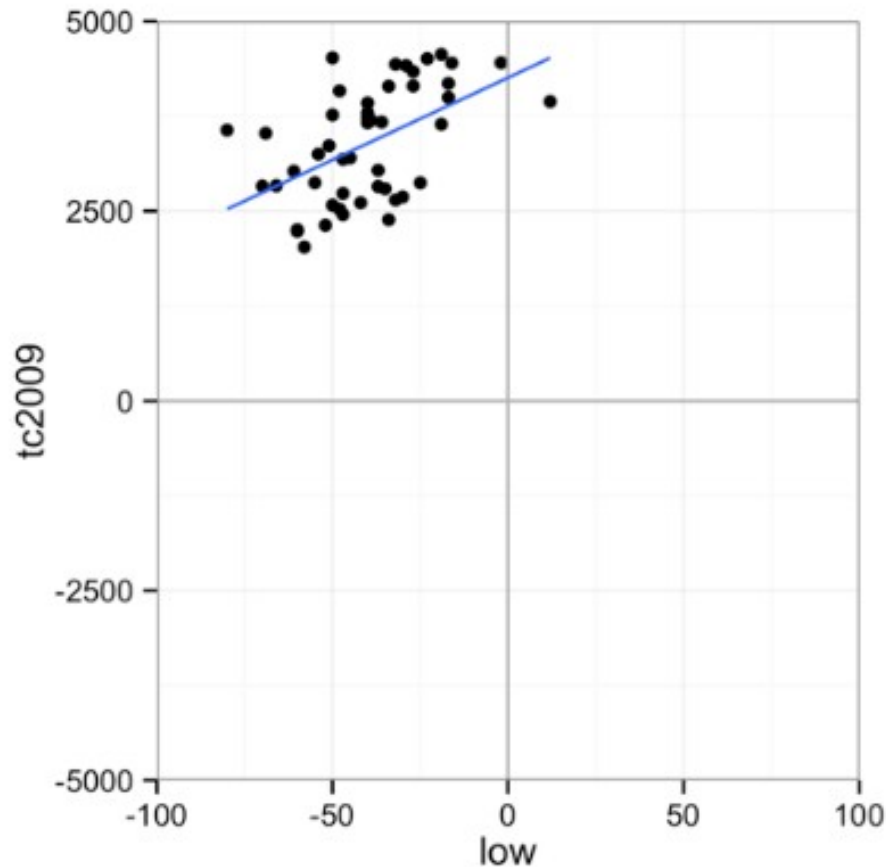
$$y = \alpha + \beta x + \epsilon$$

$$y \sim x$$

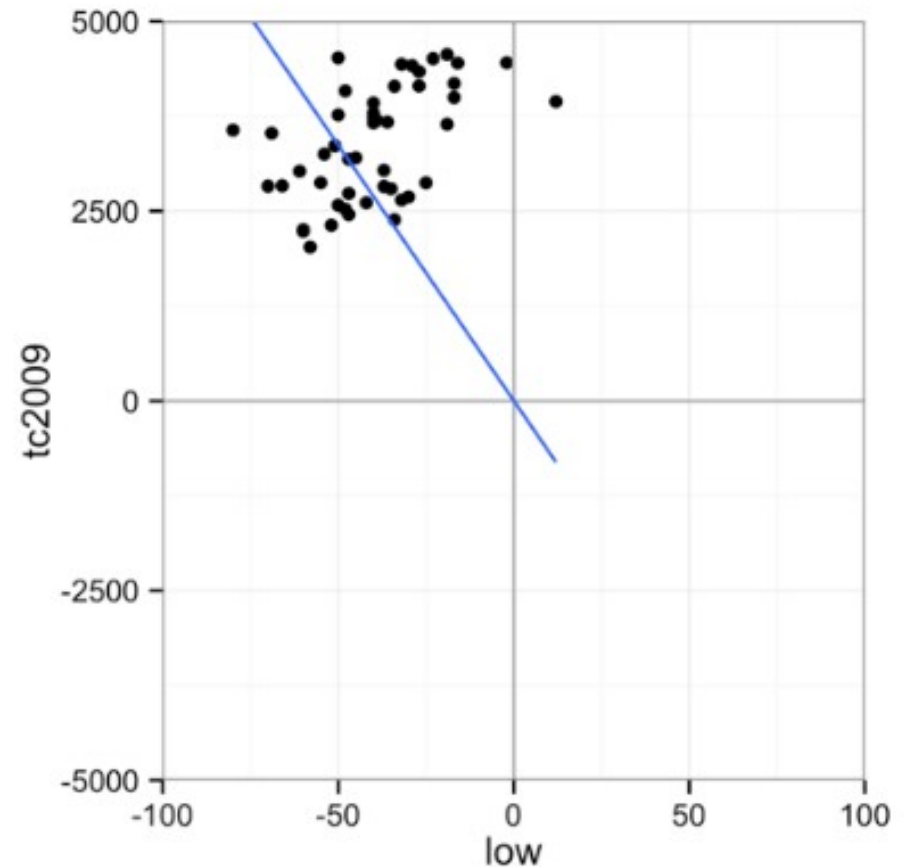


Study at $x = 0$?

(Does $x = 0$ make sense here?)



With α



Without α

Every linear model has a y intercept. Including α lets this term vary. Not including α forces the intercept to (0, 0).



Study at $x = 0$?

(Does $x = 0$ make sense here?)

- The y -intercept is the place where the regression line crosses the y -axis (where $x = 0$), and is denoted by b from $y = mx + b$
- **Meaningful interpretation:** Sometimes the y -intercept is relevant (and sometimes it is not)
- No meaning for the y -intercept when data is not present near the point where $x = 0$ (and the model suggests that data is present at this point)

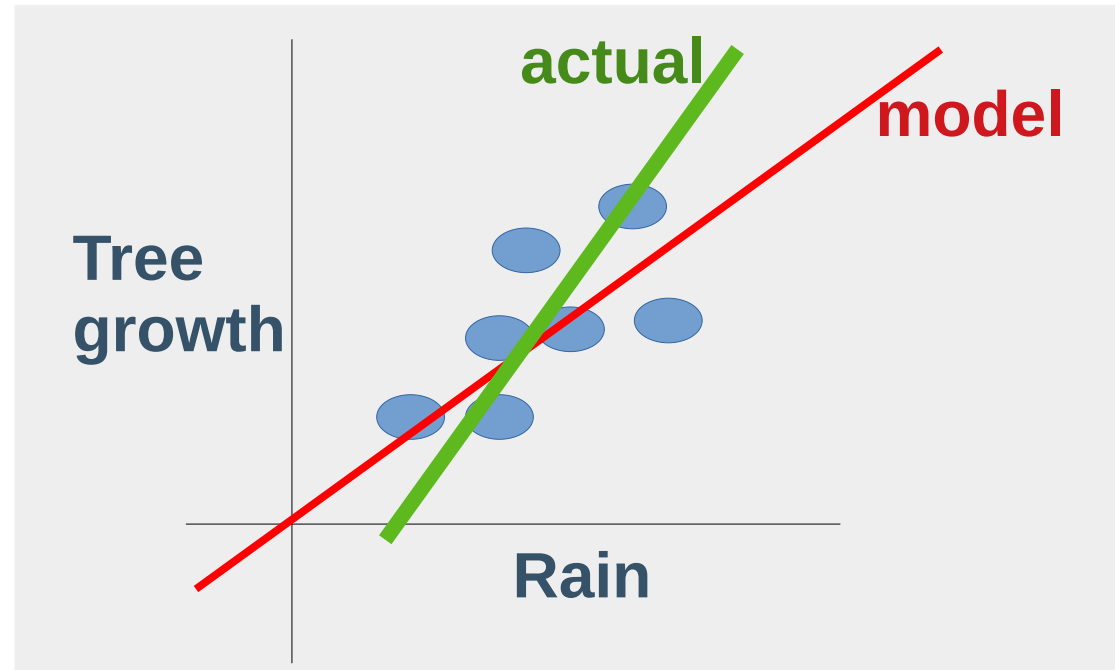


Study at $x = 0$? (Does $x = 0$ make sense here?)

Ex: A model where rain (x) is used to predict tree growth (y)

If *rain* = 0, then does *tree_growth* = 0?
Always? Trees may still grow when sunny out

Intercept not relevant:
The regression line may cross y -axis at some other point (other than zero)





An Intercept Term: To Use or Not?

FYI: You can explicitly ask for an intercept by including the number one, 1, as a formula term. You can remove the intercept by including a zero or negative 1.

equivalent - includes intercept

```
lm(tc2009 ~ 1 + low, data = crime)
```

```
lm(tc2009 ~ low, data = crime)
```

equivalent - removes intercept

```
lm(tc2009 ~ low - 1, data = crime)
```

```
lm(tc2009 ~ 0 + low, data = crime)
```



Let's Test An Intercept

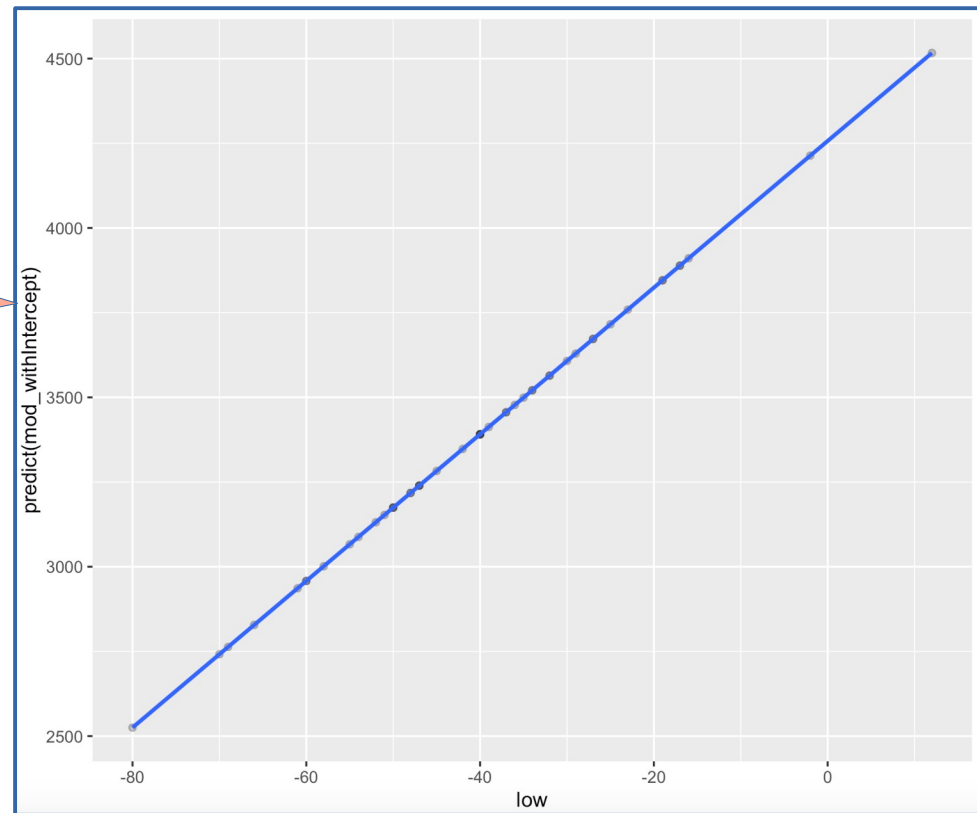
Add the intercept

```
# equivalent - includes intercept
```

```
mod_withIntercept <- lm(tc2009 ~ 1 + low, data = crime)
```

```
crime %>% ggplot(aes(x = low, y =  
predict(mod_withIntercept))) + geom_point(alpha = 1/4))  
+ geom_smooth()
```

Does this
represent
your data?





Let's Test An Intercept

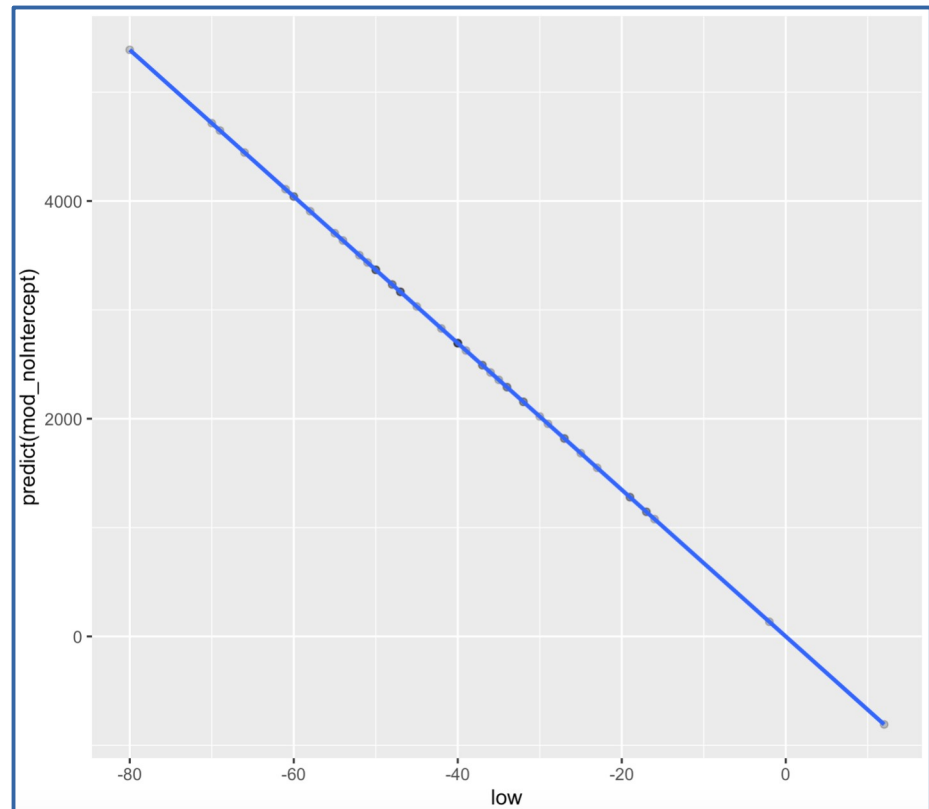
Remove the intercept

```
# equivalent - removes intercept
```

```
mod_noIntercept <- lm(tc2009 ~ low - 1, data = crime)
```

```
crime %>% ggplot(aes(x = low, y =  
predict(mod_noIntercept))) + geom_point(alpha = I(1/4)) +  
geom_smooth()
```

Does this
represent
your data?





Results: `summary(mod)`

```
> summary(mod)
```

```
Call:
```

```
lm(formula = tc2009 ~ low, data = crime)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-1134.36	-647.13	98.03	533.62	1344.30

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4256.86	233.44	18.236	< 2e-16 ***
low	21.65	5.33	4.061	0.000188 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 649.9 on 46 degrees of freedom
```

```
Multiple R-squared:  0.2639,    Adjusted R-squared:  0.2479
```

```
F-statistic: 16.49 on 1 and 46 DF,  p-value: 0.000188
```



R-squared Value

- Goodness of fit of a model: The R^2 coefficient of determination describes how well the regression predictions approximate the real data points.
- ***How well do the indep vars explain the dep var?***

$R^2 = 1$, indep variable(s) predict dep variables

$R^2 = 0$, no prediction

Residual standard error: 649.9 on 46 degrees of freedom
Multiple R-squared: 0.2639, Adjusted R-squared: 0.2479
F-statistic: 16.49 on 1 and 46 DF, p-value: 0.000188

P -Value and Model Significance

- *How do I know if this model is any good?*
- If,
 - p -value is between 0 and 0.01 (or)
 - p -value is between 0 and 0.05
- Then the model is significant
- The closer to zero, the better the model.

Residual standard error: 649.9 on 46 degrees of freedom
Multiple R-squared: 0.2639, Adjusted R-squared: 0.2479
F-statistic: 16.49 on 1 and 46 DF, p -value: 0.000188



Study the p-Value

```
#Create a simple model  
myX <- 0:100  
myY <- myX + 1  
mod <- lm(myY ~ myX)  
summary(mod)
```



Study the p-Value

```
#Create a simple model  
myX <- 0:100  
myY <- myX + 1  
mod <- lm(myY ~ myX)  
summary(mod)
```

```
Call:  
lm(formula = myY ~ myX)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-1.000e-13	-3.420e-16	1.157e-15	2.304e-15	1.242e-14

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.000e+00	2.057e-15	4.862e+14	<2e-16 ***
myX	1.000e+00	3.553e-17	2.814e+16	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.041e-14 on 99 degrees of freedom
```

```
Multiple R-squared: 1, Adjusted R-squared: 1
```

```
F-statistic: 7.921e+32 on 1 and 99 DF, p-value: < 2.2e-16
```

```
Warning message:
```

```
In summary.lm(mod) : essentially perfect fit: summary may be unreliable
```



Types of Questions to Address With Data

Q1: Is crime influenced by yearly temperature?

File: **crime.csv**



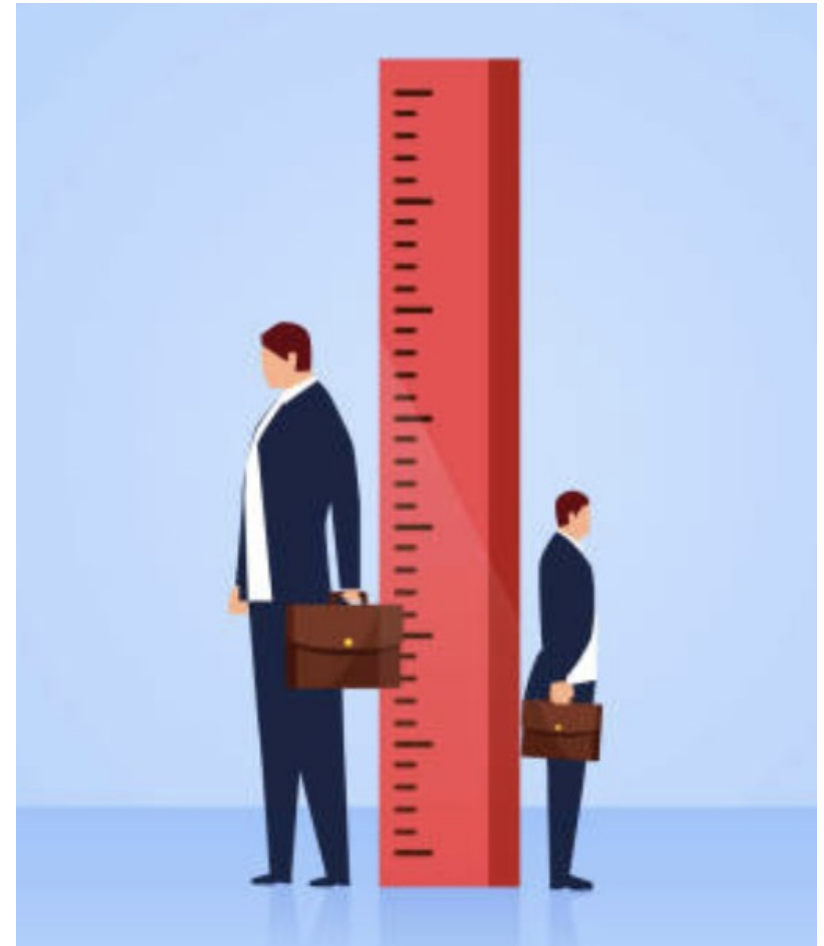
Q2: What influence is there on earning potential and personal height?

File: **wages.csv**



Consider This!

Let's try making a model with the other data set. Is there a connection between height on earning potential.





Load the Wages Data

Fit a linear model to the wages data set that predicts *earn* with *height*.

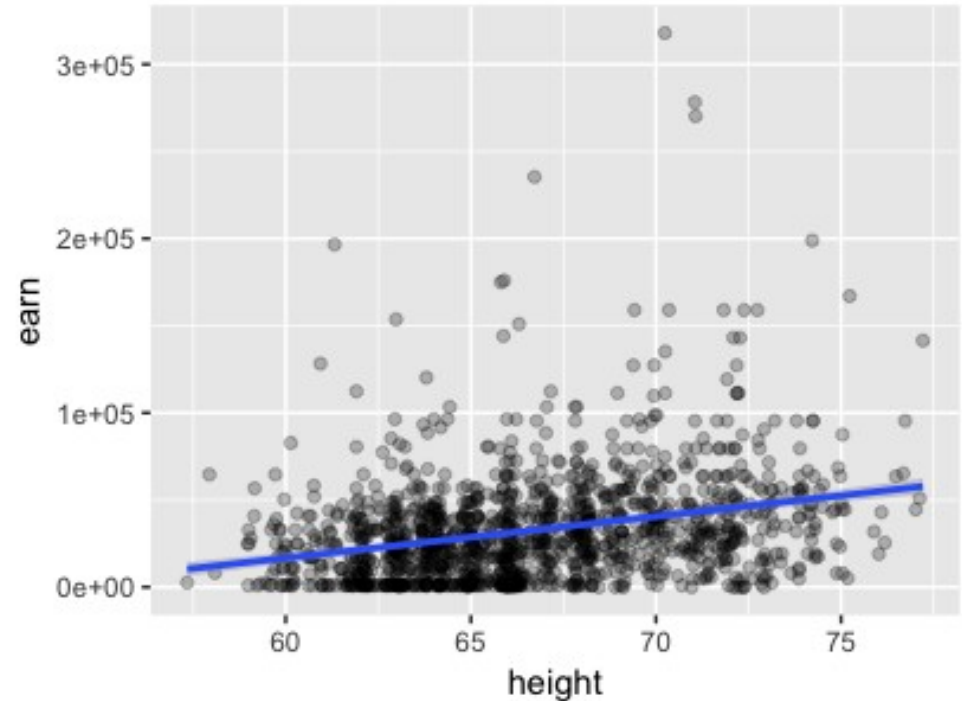
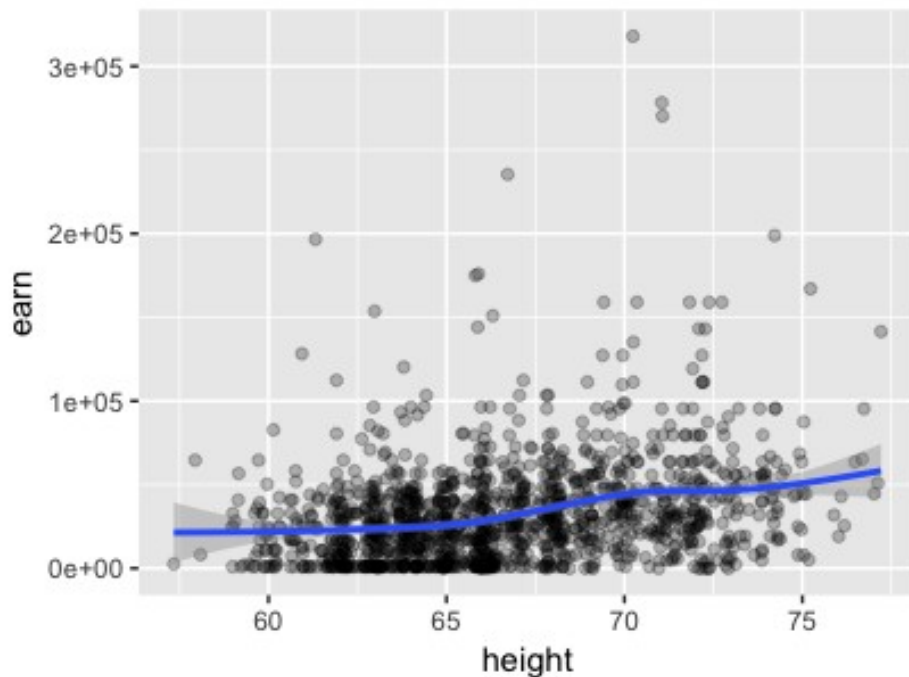
```
rm(list = ls()) # remove old vars  
# open the wages.csv dataset from  
the data.  
  
w <- file.choose() # set the  
filename  
  
wages <- read.csv(w) # load and  
read the data.
```



Do *Tall* People *Earn* More?

```
wages %>% ggplot(aes(x = height, y = earn)) +  
  geom_point(alpha = I(1/4)) + geom_smooth() # add a line  
  
wages %>% ggplot(aes(x = height, y = earn)) +  
  geom_point(alpha = I(1/4)) + geom_smooth(method = lm) #  
  linear model line
```

Try switching the x's and y's for another view.





Correlations: Earn and Height

```
# Find correlations using the "pearson"  
method
```

```
cor(wages$earn, wages$height, method =  
"pearson")
```

```
> # Find correlations using the "pearson" method  
> cor(wages$earn, wages$height, method = "pearson")  
[1] 0.2916002
```

Positive correlation

Make a Model

```
hmod <- lm(earn ~ height, data = wages)  
summary(hmod)
```

Where **dependent** var is *earn*

And **independent** var is *height*

$$\textcircled{y} = \alpha + \beta \textcircled{x} + \epsilon$$



Summary of Model

`summary(hmod)`

Build your
model's
line equation
from these
coefficients!

```
> summary(hmod)
```

```
Call:
```

```
lm(formula = wages$earn ~ wages$height)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-47903	-19744	-5184	11642	276796

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-126523	14076	-8.989	<2e-16 ***
wages\$height	2387	211	11.312	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 29910 on 1377 degrees of freedom
```

```
Multiple R-squared:  0.08503,    Adjusted R-squared:  0.08437
```

```
F-statistic:   128 on 1 and 1377 DF,  p-value: < 2.2e-16
```



Earn Regressed Over *height*

```
hmod <- lm(earn ~ height, data = wages)
```

```
coef(hmod)
```

```
## (Intercept)          height
```

```
## -126523.359      2387.196
```

$$\textit{earn} = \alpha + \beta \times \textit{height} + \epsilon$$



$$\textit{earn} = -126523.36 + 2387.20 \times \textit{height} + \epsilon$$



An Estimation

The best estimate of earn for someone 68 inches tall is

$$\text{earn} = -126523.36 + 2387.20 \times 68 + \epsilon$$

$$\text{earn} = 35806.24$$



Conclusions

Earn Regressed Over *height*

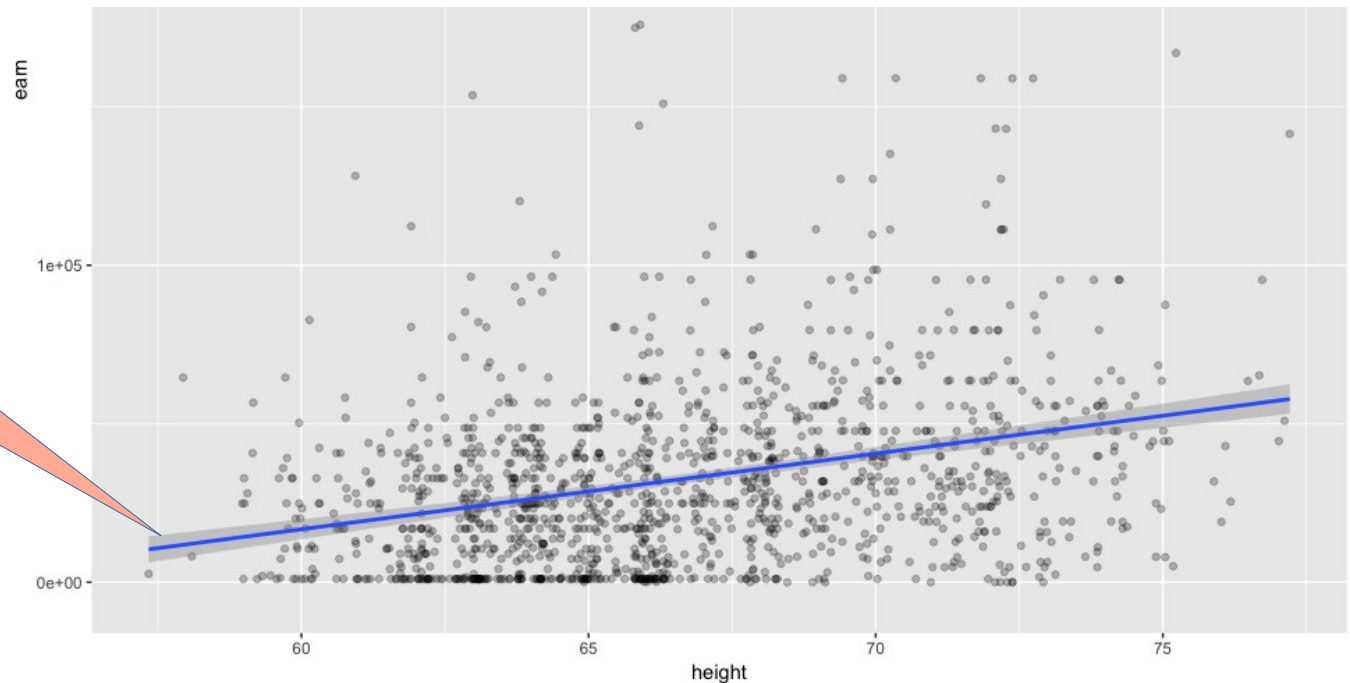
```
cor(wages$earn, wages$height, method = "pearson")
```

- Slight, but positive

```
hmod <- lm(earn ~ height, data = wages)
```

- Significant

Our model
with the
data points





Build a Model

- Fit a linear model to the wages data set
- How do we interpret the results?

Q: What happens when
we regress *earn* over *race*?

Or, How does ***race*** influence ***earn***?

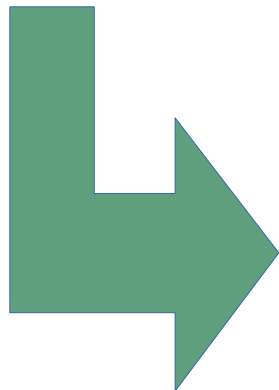


Summary

```
rmod <- lm(earn ~ race, data = wages)
coef(rmod) # get the model's y-intercepts and slopes
```

```
coef(rmod)
# (Intercept) racehispanic raceother racewhite
# 28372.09 -2886.79 3905.32 4993.33
```

summary(rmod)



Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	28372	2781	10.204	<2e-16	***
racehispanic	-2887	4515	-0.639	0.5227	
raceother	3905	6428	0.608	0.5436	
racewhite	4993	2929	1.705	0.0885	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1



Estimates From Coefficients

```
coef(rmod)
```

#	(Intercept)	racehispanic	raceother	racewhite
#	28372.09	-2886.79	3905.32	4993.33

The estimate for a white person is
 $28372.09 + 4993.33 = 33365.42$

The estimate for a other person is
 $28372.09 + 3905.32 = 32277.41$

The estimate for a hispanic person is
 $28372.09 + -2886.79 = 25485.30$

The estimate for a black person is
 $28372.09 = 28372.09$

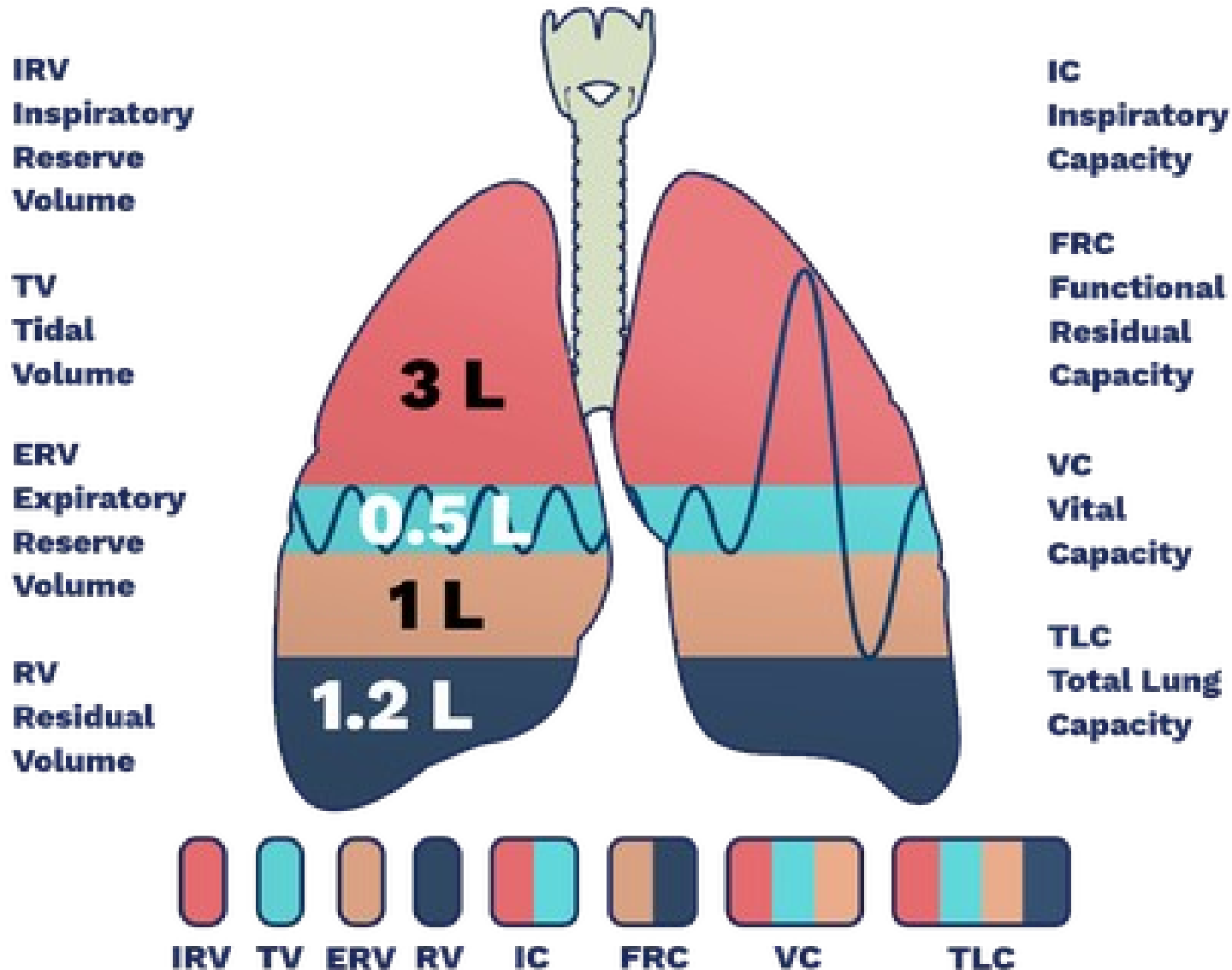


What Do Plots Also Indicate??



```
ggplot(data = wages) +  
  geom_point(mapping = aes(y = earn, x = height, color = race )) +  
  geom_smooth(mapping = aes(y = earn, x = height )) + facet_wrap(~race)
```

Lungs Capacity



File: lungCapDemo.R